

#1 · Common Occupational Skills

What does WHMIS 2015 stand for and what international system is it aligned with?

Answer

WHMIS 2015 stands for Canada's Workplace Hazardous Materials Information System. It is aligned with the Globally Harmonized System (GHS) used internationally, and it replaced the older WHMIS 1988 pictogram set and classification rules.

#2 · Common Occupational Skills

A new apprentice is about to use brazing flux for the first time. What must happen before they handle it?

Answer

The apprentice must complete WHMIS training before using any controlled product, including brazing flux, refrigerants, or cleaning solvents. This training covers pictogram recognition and hazard classification so the worker understands the risks before contact.

#3 · Common Occupational Skills

How many numbered sections does a Safety Data Sheet (SDS) have under WHMIS 2015?

Answer

A Safety Data Sheet has 16 standardized numbered sections. WHMIS 2015 standardized this format so a mechanic can locate specific technical information quickly rather than reading the entire document on site.

#4 · Common Occupational Skills

A refrigerant leak occurs on a job site. Which SDS section should the mechanic consult for accidental release measures?

Answer

Section 6 of the SDS covers accidental release measures, which applies directly to a refrigerant leak. Section 5 covers firefighting measures and section 4 covers first aid measures if exposure occurs.

#5 · Common Occupational Skills

Which SDS section lists the personal protective equipment required for a specific controlled product?

Answer

Section 8 lists exposure controls and the personal protective equipment required for that specific product. This PPE is not always the same across different refrigerants or solvents, so each SDS must be checked individually.

#6 · Common Occupational Skills

What is the difference between a pictogram and a Safety Data Sheet under WHMIS 2015?

Answer

A pictogram is a quick visual symbol on a product label showing a hazard class such as flame, corrosion, or gas cylinder. An SDS is the full 16 section technical document that gives the detailed information a pictogram alone cannot show.

#7 · Common Occupational Skills

Name three WHMIS 2015 pictograms a mechanic should be able to recognize.

Answer

A mechanic should recognize the flame pictogram, the corrosion pictogram, and the gas cylinder pictogram, along with the health hazard and exclamation mark symbols. Each pictogram represents a specific class of hazard defined under WHMIS 2015.

#8 · Common Occupational Skills

What does the signal word on an SDS label indicate, and where is it found?

Answer

The signal word, either danger or warning, indicates the severity of the hazard and is found in section 2, hazard identification. Danger indicates a more severe hazard than warning.

#9 · Common Occupational Skills

Why does liquid refrigerant expanding out of a hose pose a cold burn hazard?

Answer

Liquid refrigerant expanding rapidly out of a hose or fitting can reach temperatures well below negative 40°C almost instantly. Contact with bare skin causes frostbite-like tissue damage within seconds, so insulated, chemical-resistant gloves are mandatory.

#10 · Common Occupational Skills

A mechanic is about to recover refrigerant from a system. What PPE must they wear on their hands and face?

Answer

The mechanic must wear insulated, chemical-resistant gloves rated for refrigerant service, along with safety glasses or a full face shield. A sudden release can spray liquid refrigerant directly into the eyes, so eye protection is required for the same tasks.

#11 · Common Occupational Skills

What toxic gas can form when flux burns or refrigerant residue is heated during brazing?

Answer

Refrigerant residue left in a line and heated by the torch breaks down into toxic byproducts: chlorinated refrigerants such as R-22 can form phosgene gas, while today's common chlorine-free HFC and HFO refrigerants (R-410A, R-32, R-454B) instead form hydrofluoric acid and other toxic fluorinated compounds. A mechanic must never braze a line that has not been properly evacuated and purged of refrigerant.

#12 · Common Occupational Skills

What eye and hand protection does brazing require, and why?

Answer

Brazing requires tinted safety glasses or brazing goggles to protect against the intense light of the torch flame, and flame-resistant gloves to protect against direct heat and hot metal. This is a different set of PPE than the insulated gloves used for cold refrigerant handling.

#13 · Common Occupational Skills

A mechanic grabs an N95 particulate respirator before working near refrigerant vapour. Is this the correct choice?

Answer

No, an N95 particulate respirator does nothing against refrigerant vapour or brazing fume. The mechanic must match the respirator cartridge to the hazard identified in the SDS section 8 exposure controls, not grab whatever mask is closest.

#14 · Common Occupational Skills

List three general shop and job site PPE items besides gloves and eye protection.

Answer

Steel-toed boots are required for material handling and rooftop work, hearing protection is required around compressors and rooftop units, and hard hats are required on construction sites with overhead work. Each item addresses a distinct hazard present on a typical job site.

#15 · Common Occupational Skills

How does PPE for refrigerant handling differ from PPE for brazing?

Answer

Refrigerant handling requires insulated, chemical-resistant gloves and full eye protection against cold burns and sudden liquid release. Brazing instead requires flame-resistant gloves, tinted goggles for torch light, and ventilation or fume extraction to manage phosgene risk.

#16 · Common Occupational Skills

What three things does a confined space entry require before a worker goes in?

Answer

A confined space entry requires a written permit, atmospheric testing before and during entry for oxygen level, flammable gas, and toxic gas, and a trained attendant stationed outside who can initiate rescue procedures if needed.

#17 · Common Occupational Skills

A mechanic is about to enter a machine room with a known refrigerant leak. What is the risk if they skip atmospheric testing?

Answer

Most refrigerants are heavier than air and can pool in a low, poorly ventilated space, displacing breathable oxygen with no warning smell. Entering without testing the atmosphere risks oxygen displacement and possible asphyxiation.

#18 · Common Occupational Skills

At what height does fall protection generally apply in Ontario construction settings?

Answer

Fall protection generally applies above 3 metres in Ontario construction settings. Rooftop HVAC work often triggers fall protection requirements even at lower heights depending on the site's hazard assessment.

#19 · Common Occupational Skills

A mechanic has a full body harness on the truck but is not wearing it while working at height. Does this satisfy fall protection requirements?

Answer

No, simply having fall protection equipment present on the truck does not satisfy the requirement. The harness must be worn and properly anchored to an approved anchor point, lifeline, or guardrail system.

#20 · Common Occupational Skills

What is the 4-to-1 rule for extension ladder placement?

Answer

The 4-to-1 rule governs extension ladder angle: for every 4 units of height, the base should sit 1 unit away from the wall. This ratio keeps the ladder at a safe climbing angle.

#21 · Common Occupational Skills

What does maintaining three points of contact on a ladder mean?

Answer

Three points of contact means keeping two hands and one foot, or two feet and one hand, in contact with the ladder at all times. This applies whenever climbing or working from a ladder.

#22 · Common Occupational Skills

A mechanic needs to work near live electrical equipment and only has a metal ladder available. What should they do instead?

Answer

A metal ladder must never be used near live electrical equipment because it conducts electricity. The mechanic should use a non-conductive fiberglass ladder instead before proceeding with the work.

#23 · Common Occupational Skills

What are the first three steps a mechanic must take before applying a personal lock to a disconnect?

Answer

The mechanic must identify all energy sources feeding the equipment, notify affected workers, and shut down the equipment through its normal stopping procedure. Only after these steps does the mechanic isolate the energy source at the disconnect switch or breaker.

#24 · Common Occupational Skills

A mechanic finishes a job and finds someone else's lock still on the disconnect. Can they remove it to test the equipment?

Answer

No, only the person who applied a lock may remove it. A lock removed by anyone other than the person who placed it defeats the entire purpose of the system, because the worker under the equipment has no way of knowing their protection was removed.

#25 · Common Occupational Skills

How does a mechanic verify zero energy state after applying lockout/tagout?

Answer

The mechanic typically attempts to start the equipment through its normal controls, which should fail, and tests electrical circuits with a meter to confirm no voltage is present. Both steps confirm the energy source is truly isolated.

#26 · Common Occupational Skills

Name two forms of stored energy besides electrical charge that must be addressed during a complete lockout procedure.

Answer

A system under refrigerant pressure and a spring-loaded mechanical component both retain stored energy that must be relieved as part of a complete lockout. Stored energy is a common trap in exam scenarios because it is easy to overlook after the electrical portion is locked out.

#27 · Common Occupational Skills

A mechanic locks out a circuit but forgets a capacitor may still hold a charge. What is the danger?

Answer

A capacitor can hold a dangerous electrical charge even after the disconnect is locked out. It must be safely discharged before a mechanic works on the circuit, since the stored charge alone can cause a serious shock.

#28 · Common Occupational Skills

Does lockout/tagout apply only to electrical energy sources?

Answer

No, lockout/tagout applies to electrical, mechanical, hydraulic, and pneumatic energy alike. Any energy source capable of moving or energizing the equipment must be isolated before service work begins.

#29 · Common Occupational Skills

What is the difference between a tubing cutter and a reamer, and why are both needed?

Answer

A tubing cutter scores and cuts copper tubing cleanly with a rotating wheel, while a reamer removes the internal burr left by the cut. Skipping the reamer leaves a burr that restricts refrigerant flow and creates turbulence at that point in the circuit.

#30 · Common Occupational Skills

What is the difference between a flaring tool and a swaging tool?

Answer

A flaring tool forms a cone-shaped flare on soft copper tubing so it can join with a flare nut and fitting, commonly on service valve connections. A swaging tool instead expands the end of one tube so a second tube of the same diameter can be inserted directly for a brazed joint, eliminating a separate coupling.

#31 · Common Occupational Skills

A mechanic over-tightens a flare fitting past the manufacturer's torque specification. What happens?

Answer

Over-tightening a flare fitting cracks the flare and causes a leak. Under-tightening has the opposite problem, leaving a joint that will leak under system pressure and thermal cycling, so the correct torque value in newton-metres must always be used.

#32 · Common Occupational Skills

What four electrical properties does a digital multimeter measure?

Answer

A digital multimeter measures voltage, current, resistance, and continuity. It is used to diagnose control circuits, verify motor windings, and confirm a lockout has actually removed power.

#33 · Common Occupational Skills

What does a manifold gauge set connect to, and what does it read?

Answer

A manifold gauge set connects to a system's high side and low side service ports to read suction and discharge pressures. The mechanic then compares those readings against a pressure-temperature chart for the specific refrigerant in that system.

#34 · Common Occupational Skills

What is the difference between an electronic refrigerant leak detector and a manifold gauge set?

Answer

An electronic refrigerant leak detector senses trace refrigerant molecules in the air to find leaks too small to see or hear. A manifold gauge set instead reads suction and discharge pressure and does not detect leaks directly.

#35 · Common Occupational Skills

What is a micron gauge used for, and what is a common target vacuum level before charging?

Answer

A micron gauge measures deep vacuum far more precisely than a standard compound gauge, and is required to confirm a system has been evacuated to the target level, commonly 500 microns or lower, before charging. Moisture and non-condensable gases remain in a system evacuated only to a coarse vacuum reading.

#36 · Common Occupational Skills

A mechanic wants to check air temperature and relative humidity while commissioning a system. Which instruments do they need?

Answer

Digital thermometers and hygrometers measure air and surface temperature and relative humidity. Both are essential for calculating superheat, subcooling, and psychrometric conditions during commissioning.

#37 · Common Occupational Skills

What temperature range is typical for brazing copper tubing?

Answer

Brazing copper tubing requires temperatures around 600°C to 800°C depending on the alloy. This is more than enough to ignite nearby combustible material, so a fire watch and a clear, non-combustible work area are mandatory.

#38 · Common Occupational Skills

A mechanic finishes brazing a joint in an occupied building and packs up immediately. What safety step have they missed?

Answer

Many job sites require a dedicated fire watch person to remain at the location for a defined period after brazing is complete. Smoldering material can ignite well after the torch is shut off, so leaving immediately skips this required step.

#39 · Common Occupational Skills

Why is nitrogen purging used during brazing?

Answer

Dry nitrogen flows through the tubing at a low, controlled pressure while the joint is heated, displacing oxygen inside the tube and preventing copper oxide scale from forming on the interior wall. That scale, if allowed to form, breaks loose and clogs filter driers, expansion devices, and compressor valves.

#40 · Common Occupational Skills

How should a mechanic set nitrogen flow rate during a brazing operation?

Answer

Nitrogen flow should be low enough that it does not blow out the brazing filler material but high enough to maintain a positive, oxygen-free atmosphere inside the tube for the duration of the braze. Both extremes create a problem, either scale formation or a disrupted joint.

#41 · Common Occupational Skills

A mechanic is about to braze a line but the system still contains refrigerant. What is the direct life safety issue?

Answer

Refrigerant contacting the torch flame breaks down into toxic byproducts, phosgene gas for chlorinated refrigerants like R-22 and hydrofluoric acid and other toxic fluorinated gases for today's common chlorine-free HFC/HFO refrigerants. Proper evacuation and recovery before cutting into or brazing a line is both an environmental requirement and a direct life safety issue, not simply good practice.

#42 · Common Occupational Skills

How does sling angle affect the load carried by each leg of a rigging setup?

Answer

A sling rigged at a shallow angle from horizontal carries dramatically more tension than the same sling rigged closer to vertical. Exceeding a sling's rated capacity at a poor angle is a common cause of rigging failure.

#43 · Common Occupational Skills

What must a mechanic check on rigging components like slings, shackles, and hooks before use?

Answer

Every rigging component has a working load limit stamped or tagged on it, and the mechanic must inspect it for wear, cuts, deformation, or missing tags before use. Anything damaged must be removed from service immediately.

#44 · Common Occupational Skills

A mechanic is directing a rooftop unit lift but cannot use a radio to communicate with the crane operator. What should they use instead?

Answer

Standard hand signals are used to communicate with a crane operator when radio communication is not practical. The mechanic must know these signals and stay clear of the load's path and swing radius at all times.

#45 · Common Occupational Skills

What is proper manual lifting technique for handling smaller equipment components?

Answer

Proper technique means bending at the knees rather than the back, keeping the load close to the body, and asking for a second person on an awkward or heavy item. This prevents the soft tissue injuries that account for a large share of lost time incidents in this trade.

#46 · Common Occupational Skills

What does a circle with an X typically represent on a mechanical drawing?

Answer

A circle with an X often marks a fan or motor on a mechanical drawing. Parallel lines with an airflow arrow represent ductwork, and specific line weights distinguish supply, return, and exhaust systems from one another.

#47 · Common Occupational Skills

What is the difference between a work order and a service report?

Answer

A work order describes the job requested and any site-specific access or safety information the mechanic needs before arriving. A service report is completed after the call and documents the equipment serviced, diagnosis, work performed, parts used, and readings taken.

#48 · Common Occupational Skills

Why is refrigerant tracking documentation required on every service call?

Answer

Refrigerant tracking documentation, required under federal and provincial regulation, records the type and quantity of refrigerant added to or recovered from a system on every service call. Inaccurate or missing records can result in regulatory penalties for the contractor.

#49 · Common Occupational Skills

A worker believes a task is likely to endanger them but their supervisor insists it be done anyway. What are the worker's rights under Ontario's OHSA?

Answer

A worker who has reason to believe a task is likely to endanger themselves or another worker can refuse unsafe work. The employer must investigate the refusal, and the worker cannot be disciplined for a good-faith refusal made through the proper process.

#50 · Common Occupational Skills

What is CSA B52, and what does a Red Seal endorsement certify?

Answer

CSA B52 is the Mechanical Refrigeration Code, setting design, installation, and safety requirements for refrigeration systems in Canada, including machine room ventilation and refrigerant classification limits. A Red Seal endorsement certifies that a tradesperson's skills meet a nationally standardized benchmark, allowing them to work anywhere in Canada without re-testing in each province.

#51 · Routine Trade Activities

What is the key difference between heat and temperature in refrigeration theory?

Answer

Temperature measures molecular activity, how fast molecules vibrate. Heat is energy in transit, measured in joules or kilojoules, and always flows from a warmer object to a cooler one. Confusing the two is a common early mistake for apprentices.

#52 · Routine Trade Activities

Define specific heat and state water's approximate specific heat value.

Answer

Specific heat is the energy needed to raise one kilogram of a substance by one degree Celsius, expressed in $\text{kJ}/(\text{kg}\cdot^{\circ}\text{C})$. Water has a very high specific heat, about $4.19 \text{ kJ}/(\text{kg}\cdot^{\circ}\text{C})$, which is why it is used as a heat transfer medium in chilled water and hydronic systems.

#53 · Routine Trade Activities

A block of steel and a glass of water are both at 40°C. Do they hold the same amount of heat energy?

Answer

No. Even though they are at the same temperature, steel and water have different specific heat values, so they store different amounts of heat energy. Temperature tells you molecular activity, not the total energy stored in a material.

#54 · Routine Trade Activities

Explain why a mechanical refrigeration system is best described as heat relocation rather than cold creation.

Answer

A refrigeration system does not create cold. It absorbs heat from a space where it is unwanted, such as a walk-in cooler, and rejects that same heat somewhere else, typically outdoors. Treating heat as something moved rather than destroyed is the foundation for understanding the entire cycle.

#55 · Routine Trade Activities

Why does a mechanic need specific heat values for air, refrigerant, and metals on the job?

Answer

Specific heat values are used directly when calculating heat loads or working out how much energy a coil is removing from an airstream. Each material, air, refrigerant, or metal, stores and releases heat differently, so the correct value must be used for an accurate calculation.

#56 · Routine Trade Activities

What is sensible heat, and give an example from an air conditioning system?

Answer

Sensible heat changes a substance's temperature without changing its physical state. Heating air from 18°C to 24°C is an example, and that temperature rise can be measured directly with a thermometer.

#57 · Routine Trade Activities

Define latent heat and explain when it is absorbed or released.

Answer

Latent heat is the heat absorbed or released when a substance changes state, from liquid to vapour or vapour back to liquid, at a constant temperature. No temperature change occurs while latent heat is being transferred, only a change of state.

#58 · Routine Trade Activities

A pot of water boils at sea level and stays at 100°C the whole time heat is added. Why doesn't the temperature keep rising?

Answer

The added heat is changing the water from liquid to vapour rather than raising its temperature further, since boiling is a latent heat process. This is the same principle that governs refrigerant boiling inside an evaporator at a constant saturation temperature.

#59 · Routine Trade Activities

What kind of heat transfer occurs inside an evaporator as refrigerant boils?

Answer

Liquid refrigerant boils at a low, constant saturation temperature and absorbs a large quantity of latent heat as it turns to vapour. No sensible temperature rise occurs in the refrigerant while it is boiling.

#60 · Routine Trade Activities

Compare the latent heat process happening in the evaporator with the one happening in the condenser.

Answer

In the evaporator, refrigerant absorbs latent heat and boils from liquid to vapour at a low, constant saturation temperature. In the condenser, refrigerant releases that latent heat and condenses from vapour back to liquid at a higher, constant saturation temperature.

#61 · Routine Trade Activities

List the four stages of the vapour-compression refrigeration cycle in order.

Answer

The four stages are compression, condensation, expansion, and evaporation. Every basic refrigeration and air conditioning system, from a household fridge to a large rooftop unit, runs this same cycle.

#62 · Routine Trade Activities

What happens to refrigerant during the compression stage, and which component performs it?

Answer

Low-pressure, low-temperature vapour enters the compressor from the suction line and is mechanically squeezed into a much smaller volume. This raises pressure and temperature significantly, producing hot, high-pressure superheated vapour at the discharge line.

#63 · Routine Trade Activities

Describe what happens to refrigerant as it passes through the condenser.

Answer

Hot discharge vapour gives up heat to a cooler medium, usually outdoor air or, in commercial systems, water. It first cools to its saturation temperature, then condenses into high-pressure liquid, and should leave the condenser as subcooled liquid.

#64 · Routine Trade Activities

What state should refrigerant be in as it exits the condenser on a properly operating system?

Answer

Refrigerant should exit the condenser as subcooled liquid, meaning its temperature has dropped slightly below the saturation temperature for that pressure. This confirms the condenser fully rejected the latent heat needed to completely liquefy the refrigerant.

#65 · Routine Trade Activities

Explain what happens to high-pressure liquid refrigerant as it passes through the metering device.

Answer

High-pressure liquid is forced through a small restriction, such as a TXV, EXV, or fixed orifice, and its pressure drops sharply on the other side. That drop flashes a portion of the liquid into vapour instantly, cooling the remaining liquid down to the low evaporator temperature.

#66 · Routine Trade Activities

What is the physical state of refrigerant as it leaves the metering device and enters the evaporator?

Answer

Refrigerant leaving the metering device is a low-pressure mixture of liquid and vapour. As that mixture moves through the evaporator, the remaining liquid boils off at the low evaporator saturation pressure.

#67 · Routine Trade Activities

By the time refrigerant leaves the evaporator, what condition should it be in and why does that matter?

Answer

All the liquid should be gone and the vapour should be slightly superheated. This confirms complete evaporation occurred before the vapour re-enters the compressor, protecting the compressor from liquid slugging damage.

#68 · Routine Trade Activities

What does a pressure-temperature (P-T) chart show for a given refrigerant?

Answer

A P-T chart shows the fixed relationship between pressure and the temperature at which a refrigerant boils or condenses. For a given refrigerant, one specific pressure corresponds to exactly one specific boiling or condensing temperature, as long as liquid and vapour are present together.

#69 · Routine Trade Activities

Define saturation as it applies to refrigerant in an evaporator or condenser.

Answer

Saturation is the condition where liquid and vapour coexist at a pressure and temperature matching the P-T chart exactly. Refrigerant is at saturation while boiling in the evaporator and while condensing in the condenser.

#70 · Routine Trade Activities

Why does raising head pressure raise the condensing temperature of a system?

Answer

Saturation temperature rises as pressure rises, so increasing head pressure directly increases the temperature at which refrigerant condenses. This same relationship explains why lowering suction pressure produces a colder evaporator.

#71 · Routine Trade Activities

Describe the correct process a technician follows to interpret a pressure gauge reading using saturation.

Answer

The technician reads a pressure, converts it to a saturation temperature using the P-T chart for that specific refrigerant, and only then compares that number to actual measured line temperatures. Reading gauges without applying this saturation step is close to useless for diagnosis.

#72 · Routine Trade Activities

What can go wrong if a technician uses the wrong P-T chart or forgets which refrigerant is in a system?

Answer

Using the wrong chart, or forgetting which refrigerant is in a system, leads directly to a wrong diagnosis. This mistake can result in dangerous overcharging or undercharging of the system.

#73 · Routine Trade Activities

Define superheat and state which condition must be true for it to apply.

Answer

Superheat is the difference between the actual temperature of refrigerant vapour and the saturation temperature at the pressure it is measured at. It only applies once refrigerant has fully evaporated with no liquid remaining.

#74 · Routine Trade Activities

State the exact formula a technician uses to calculate superheat.

Answer

Superheat equals actual line temperature minus saturation temperature at the measured pressure. The saturation temperature is found by converting the measured pressure using the refrigerant's P-T chart.

#75 · Routine Trade Activities

Describe the exact steps a technician follows to measure superheat in the field.

Answer

The technician reads the low-side pressure, converts it to a saturation temperature using the P-T chart, and measures the actual line temperature at the same point with a clamp-on thermocouple. Subtracting the saturation temperature from the actual line temperature gives the superheat value.

#76 · Routine Trade Activities

Worked calculation: an R-410A suction pressure corresponds to 6°C saturation, and the measured line temperature is 14°C. Find the superheat.

Answer

Superheat = actual line temperature – saturation temperature = $14^{\circ}\text{C} - 6^{\circ}\text{C} = 8^{\circ}\text{C}$. An 8°C superheat on a fixed-orifice system near design conditions is a reasonable, healthy reading.

#77 · Routine Trade Activities

Worked calculation: saturation temperature at the suction pressure is negative 8°C, and the actual suction line temperature is negative 1°C. Find the superheat.

Answer

Superheat = actual line temperature – saturation temperature = $-1^{\circ}\text{C} - (-8^{\circ}\text{C}) = 7^{\circ}\text{C}$. A 7°C superheat is a typical, healthy field reading confirming the evaporator fully boiled off its refrigerant.

#78 · Routine Trade Activities

What does a superheat reading near 0°C warn a technician about?

Answer

Near-zero superheat warns that liquid refrigerant may be reaching the compressor. This can wash out the crankcase oil or cause slugging damage to the compressor.

#79 · Routine Trade Activities

What does an unusually high superheat reading suggest is happening in the system?

Answer

High superheat suggests the evaporator is starving for refrigerant. This points to undercharge, a restriction in the refrigerant piping, or a metering device that is not feeding enough refrigerant.

#80 · Routine Trade Activities

Define subcooling and state which condition must be true for it to apply.

Answer

Subcooling is a measurement taken on the liquid side of the system, and it only applies once refrigerant has fully condensed. Subcooling proves the condenser is rejecting enough heat to fully liquefy the refrigerant before it reaches the metering device.

#81 · Routine Trade Activities

State the exact formula a technician uses to calculate subcooling.

Answer

Subcooling equals saturation temperature at the measured pressure minus actual liquid line temperature. This is the reverse order from the superheat formula, since subcooling measures how far below saturation the liquid has dropped.

#82 · Routine Trade Activities

Worked calculation: an R-410A high side reads 2800 kPa, corresponding to a saturation temperature of 48°C, and the liquid line reads 38°C. Find the subcooling.

Answer

Subcooling = saturation temperature – actual liquid line temperature = $48^{\circ}\text{C} - 38^{\circ}\text{C} = 10^{\circ}\text{C}$. A subcooling reading in this range is common for a properly charged TXV system, though the technician should always confirm the target against the specific manufacturer's specification.

#83 · Routine Trade Activities

Compare what low subcooling and high subcooling each indicate about a system.

Answer

Low subcooling points to an undercharged system or a condenser not rejecting enough heat. High subcooling points to an overcharged system, a liquid line restriction, or a condenser oversized or exposed to too much cold ambient air.

#84 · Routine Trade Activities

Compare how charging method differs between a TXV system and a fixed-orifice system.

Answer

On a TXV system, charging is set by target subcooling, since the TXV actively controls superheat itself. On a fixed-orifice system, charging is set by target superheat instead, because a fixed orifice cannot self-adjust the way a TXV can.

#85 · Routine Trade Activities

What refrigerant family became the industry standard after older ozone-depleting refrigerants were phased out?

Answer

Hydrofluorocarbons (HFCs) became the industry standard after older ozone-depleting refrigerants were phased out under the Montreal Protocol. HFCs do not damage the ozone layer, but many have a high global warming potential.

#86 · Routine Trade Activities

Name three common HFC refrigerants still widely serviced in Canada and one typical application for each.

Answer

R-134a is used in chillers, transport refrigeration, and automotive systems. R-410A is used in residential and light commercial air conditioning, and R-404A is used in commercial refrigeration and freezers.

#87 · Routine Trade Activities

Which lower-GWP refrigerant is used as an alternative to R-410A, and which is the blended replacement for R-410A in new equipment?

Answer

R-32 is used as a lower-GWP alternative to R-410A. R-454B is a blended HFO/HFC replacement for R-410A specifically designed for new equipment.

#88 · Routine Trade Activities

What refrigerant is the standard replacement for R-134a in automotive air conditioning?

Answer

R-1234yf is the standard replacement for R-134a in automotive air conditioning. It is a hydrofluoroolefin (HFO) with a global warming potential close to 1, far lower than R-134a.

#89 · Routine Trade Activities

Define ozone depletion potential (ODP) and state what current HFCs and HFOs have in common regarding it.

Answer

ODP rates how damaging a refrigerant is to the stratospheric ozone layer, using R-11 as the reference point at 1.0. All HFCs and HFOs in current use have an ODP of zero, since they contain no chlorine.

#90 · Routine Trade Activities

Define global warming potential (GWP) and state the reference value used for carbon dioxide.

Answer

GWP rates how much a given mass of a refrigerant contributes to atmospheric warming compared to the same mass of carbon dioxide over 100 years. Carbon dioxide is set as the reference point at a GWP of 1.

#91 · Routine Trade Activities

Compare the approximate GWP values of R-404A and R-32.

Answer

R-404A has a very high GWP, over 3,900, which is why it is being phased down aggressively. R-32 sits much lower, around 675, roughly one sixth of R-404A's value.

#92 · Routine Trade Activities

What flammability consideration does a mechanic need to weigh when comparing several newer lower-GWP refrigerants for a retrofit?

Answer

Several newer lower-GWP options, such as R-32, R-454B, and R-1234yf, carry a mild flammability rating known as A2L, which older HFCs did not have. A mechanic weighs GWP alongside performance, cost, and this flammability rating when comparing refrigerants for a retrofit.

#93 · Routine Trade Activities

What is an azeotropic blend, and give an example?

Answer

An azeotropic blend behaves like a single substance, boiling and condensing at one consistent temperature at a given pressure with no practical glide. R-410A is an example of an azeotropic blend.

#94 · Routine Trade Activities

What is a zeotropic blend, and give two examples?

Answer

A zeotropic blend mixes refrigerants with different boiling points and exhibits glide, boiling and condensing over a range of temperatures rather than at one fixed point. R-404A and R-407C are both examples of zeotropic blends.

#95 · Routine Trade Activities

Explain what bubble point and dew point mean on a P-T chart for a zeotropic blend.

Answer

A zeotropic blend's P-T chart lists both a bubble point, where the last drop of liquid vaporizes, and a dew point, where the last drop of vapour condenses, at each pressure. This range between the two points is the glide.

#96 · Routine Trade Activities

Why must zeotropic blends always be charged into a system as a liquid rather than a vapour?

Answer

Pulling vapour off the top of a zeotropic blend cylinder would draw off the more volatile component first, shifting the blend's composition in both the cylinder and the system. To avoid this fractionation, zeotropic blends must always be charged as a liquid, never as a vapour.

#97 · Routine Trade Activities

What does the British thermal unit (BTU) measure, and where does it still commonly appear in the trade?

Answer

The BTU measures heat energy. BTU per hour (BTU/h) still appears constantly on nameplates and older documentation, especially on equipment shared across the Canada-US border, even as the industry shifts toward SI units.

#98 · Routine Trade Activities

Define one ton of refrigeration and give its two standard conversion values.

Answer

One ton equals the heat required to melt one short ton (2,000 lb) of ice in 24 hours. It converts to 3.517 kW, or roughly 12,000 BTU/h.

#99 · Routine Trade Activities

Worked calculation: convert a 3 ton cooling capacity to kilowatts.

Answer

$3 \text{ tons} \times 3.517 \text{ kW per ton} = 10.551 \text{ kW}$. A 3 ton unit therefore has a rated cooling capacity of about 10.55 kW.

#100 · Routine Trade Activities

Worked calculation: a rooftop unit is rated at 17.6 kW. About how many tons is that?

Answer

$17.6 \text{ kW} \div 3.517 \text{ kW per ton} \approx 5.0 \text{ tons}$. This matches a typical five ton rooftop unit's rated cooling capacity.

#101 • Routine Trade Activities

Worked calculation: convert 24,000 BTU/h of cooling capacity into kilowatts.

Answer

24,000 BTU/h equals 2 tons of refrigeration, since one ton equals 12,000 BTU/h. $2 \text{ tons} \times 3.517 \text{ kW per ton} = 7.034 \text{ kW}$.

#102 • Routine Trade Activities

Define coefficient of performance (COP) and state its formula.

Answer

COP is a dimensionless efficiency ratio comparing useful output to energy input, both measured in the same units. $\text{COP} = \text{useful output} \div \text{energy input}$, and a higher COP means more cooling per unit of energy consumed.

#103 • Routine Trade Activities

Worked calculation: a system delivers 10 kW of useful cooling output while consuming 2.5 kW of energy input. What is its COP?

Answer

$\text{COP} = \text{useful output} \div \text{energy input} = 10 \text{ kW} \div 2.5 \text{ kW} = 4$. A COP of 4 means the system delivers four units of cooling for every unit of energy it consumes.

#104 • Routine Trade Activities

Define energy efficiency ratio (EER) and how it differs from COP in its units.

Answer

EER is BTU/h of cooling divided by watts of electrical input, measured at one specified outdoor condition. Unlike COP, which is dimensionless because both sides use the same units, EER mixes BTU/h and watts, so it carries mixed units.

#105 • Routine Trade Activities

Worked calculation: a unit produces 12,000 BTU/h of cooling while drawing 1,500 watts of electrical input. What is its EER?

Answer

$\text{EER} = \text{BTU/h of cooling} \div \text{watts of input} = 12,000 \div 1,500 = 8$. An EER of 8 describes the unit's efficiency at that one specified outdoor condition.

#106 • Routine Trade Activities

What does SEER (or SEER2) measure, and why is it the number most commonly quoted for residential air conditioning in Canada?

Answer

SEER, or SEER2 under the newer standard, measures the same idea as EER but averaged across a full cooling season rather than one fixed condition. It is the number most commonly quoted for residential air conditioning sold in Canada, and provincial codes increasingly set minimum SEER2 thresholds for new installs.

#107 · Routine Trade Activities

What do the vertical and horizontal axes of a pressure-enthalpy (P-h) diagram represent?

Answer

The vertical axis of a P-h diagram represents pressure, and the horizontal axis represents enthalpy, the refrigerant's total heat content per kilogram. A curved dome-shaped line runs through the middle of the chart marking the saturation boundary.

#108 · Routine Trade Activities

What do the regions to the left of the dome, to the right of the dome, and inside the dome represent on a P-h diagram?

Answer

To the left of the dome is subcooled liquid, and to the right is superheated vapour. Inside the dome is the two-phase region, where liquid and vapour coexist at saturation.

#109 • Routine Trade Activities

How does the compression stage appear when plotted on a P-h diagram?

Answer

Compression appears as a nearly vertical line moving up and slightly right, from low-pressure vapour into high-pressure superheated vapour. Enthalpy rises during compression because the compressor adds mechanical work to the refrigerant.

#110 • Routine Trade Activities

How does the condensation stage appear when plotted on a P-h diagram?

Answer

Condensation appears as a horizontal line moving left at constant high pressure, crossing through the dome into subcooled liquid. Enthalpy drops throughout condensation as heat is rejected to the outdoor air or water.

#111 • Routine Trade Activities

How do the expansion and evaporation stages each appear when plotted on a P-h diagram?

Answer

Expansion appears as a straight vertical drop at constant enthalpy, since a metering device transfers no heat and does no work, only dropping pressure. Evaporation appears as a horizontal line moving right at constant low pressure, with enthalpy rising as the refrigerant absorbs heat and crosses back through the dome into superheated vapour.

#112 • Routine Trade Activities

Define conduction and give an example from inside a refrigeration coil.

Answer

Conduction is heat transfer through direct contact within or between solid materials. Heat moving through the wall of a copper evaporator tube, from the refrigerant side to the fin side, is an example of conduction.

#113 · Routine Trade Activities

Define convection and explain why refrigeration equipment depends on it so heavily.

Answer

Convection is heat transfer between a solid surface and a moving fluid, such as air blowing across a condenser coil. Refrigeration equipment depends heavily on forced convection from fans and pumps to move heat efficiently across coils.

#114 · Routine Trade Activities

Define radiation as a heat transfer mechanism and explain its practical effect on equipment placement.

Answer

Radiation is heat transfer through electromagnetic waves with no physical medium required, such as heat radiating off a hot compressor shell. Although radiation plays a smaller role in most refrigeration heat exchange, it still affects equipment placement, particularly for outdoor units in direct sun.

#115 • Routine Trade Activities

Compare conduction and convection by identifying which mechanism dominates inside a copper coil wall versus across its outer surface.

Answer

Heat moving through the wall of a copper tube from refrigerant to fins is conduction, transfer through direct contact within a solid. Air blowing across the coil surface to remove that heat is convection, transfer between a solid surface and a moving fluid.

#116 • Routine Trade Activities

What does psychrometrics study, and in which application does it matter most?

Answer

Psychrometrics is the study of moist air properties. It matters most in air conditioning and comfort cooling, and less so in straight refrigeration work.

#117 · Routine Trade Activities

Define dry bulb temperature and wet bulb temperature, and state how their readings compare to each other.

Answer

Dry bulb temperature is the ordinary air temperature read on a standard thermometer. Wet bulb temperature is read using a thermometer with a wetted wick exposed to airflow, and it always reads at or below dry bulb, except at 100 percent relative humidity, where the two are equal.

#118 · Routine Trade Activities

Define relative humidity and state what it is measured as a ratio of.

Answer

Relative humidity is the ratio of the actual moisture content of air to the maximum moisture it could hold at that same temperature, expressed as a percentage. It is one of the core properties read from a psychrometric chart.

#119 · Routine Trade Activities

Define dew point and explain what happens on an evaporator coil running colder than the entering air's dew point.

Answer

Dew point is the temperature at which air becomes fully saturated and moisture begins condensing out. An evaporator coil running colder than the entering air's dew point produces condensate, which is exactly what drains off through the condensate line.

#120 · Routine Trade Activities

What readings does a mechanic combine with a psychrometric chart to find relative humidity, dew point, and total heat content of an airstream?

Answer

A mechanic uses wet bulb and dry bulb temperature readings together with a psychrometric chart. From those two readings alone, the chart yields relative humidity, dew point, and the total heat content of the airstream.

#121 • Routine Trade Activities

Why must refrigerant oil be miscible with the refrigerant it circulates with?

Answer

Refrigerant oil circulates with the refrigerant to lubricate the compressor's moving parts, so the two must be miscible, mixing and travelling together through the piping. Choosing the correct oil for the refrigerant in use is not optional.

#122 • Routine Trade Activities

Why is mineral oil generally not compatible with modern HFC refrigerants?

Answer

Mineral oil was the traditional lubricant used with older CFC and HCFC refrigerants like R-12 and R-22. It does not mix properly with most modern HFCs, and can separate out, collecting in the evaporator and starving the compressor of lubrication.

#123 · Routine Trade Activities

What type of oil is the standard lubricant for most HFC refrigerants today, and why is its hygroscopic nature a concern?

Answer

Polyol ester (POE) oil is the standard lubricant for most HFC refrigerants today, including R-410A, R-134a, and R-404A, because it stays miscible across the system's full operating range. POE is also highly hygroscopic, readily absorbing moisture from the air, so containers must stay sealed and system exposure to open air minimized during service.

#124 · Routine Trade Activities

What is alkylbenzene (AB) oil used for, and is it recommended as a primary lubricant for HFC systems?

Answer

Alkylbenzene oil is a synthetic oil sometimes used with HCFC refrigerants like R-22, particularly in low-temperature work, and occasionally blended with mineral oil to improve miscibility in retrofits. It is not generally recommended as a primary lubricant for HFC systems.

#125 • Routine Trade Activities

What oil type do newer lower-GWP refrigerants like R-32, R-454B, and R-1234yf most commonly use?

Answer

R-32 and R-454B, used in stationary equipment, most commonly use POE oils. R-1234yf is the exception: since it is used almost exclusively in automotive air conditioning, it is typically serviced with PAG oil, the same family used with automotive R-134a systems, except on hybrid and electric-vehicle compressors, which require POE oil for its non-conductive properties. A mechanic should always confirm the specific recommendation from the equipment and refrigerant manufacturers, since compatibility differs by application and blend.

#126 • Installation Planning

What are the two separate components of a cooling load, and why must each be calculated independently?

Answer

Sensible heat changes air temperature without changing moisture content, driven by solar gain, lighting, equipment and conducted heat. Latent heat is the energy needed to condense water vapour out of the air stream, driven by occupant moisture and outside air infiltration. They are calculated separately because a system sized only on total capacity can still fail to control humidity if the equipment's sensible heat ratio does not match the building's.

#127 • Installation Planning

A restaurant's new AC keeps the space cold but clammy and cycles off quickly. What design mistake likely caused this?

Answer

The equipment's sensible heat ratio (SHR) probably does not match the actual building load's SHR, often because a generic per-square-metre rule of thumb was used instead of a proper load calculation. The unit satisfies the thermostat on sensible cooling and short cycles before it runs long enough to remove the high latent load from occupants and cooking. The fix is a proper load calculation that sizes to the space's real SHR, not just total tonnage.

#128 • Installation Planning

List three typical sources of sensible heat load in a building.

Answer

Solar gain through windows, heat given off by lighting and equipment, and heat conducted through walls, roofs and floors are all sensible sources. Occupants' body heat is also sensible. None of these add moisture to the air, they only raise dry bulb temperature.

#129 • Installation Planning

List three typical sources of latent heat load in a building.

Answer

Occupant respiration and perspiration, moisture infiltration from outside air, and process moisture from cooking or manufacturing are all latent sources. A space with many occupants, such as a restaurant or gym, can carry a very high latent load even with a modest sensible load. Latent load is the energy needed to condense that moisture out at the evaporator coil.

#130 • Installation Planning

Why does using Vancouver's outdoor design temperature for a Winnipeg load calculation produce a wrong result?

Answer

Design conditions are specific to the local climate, and Winnipeg's outdoor design temperature and humidity are very different from Vancouver's. Using the wrong city's design condition throws off the whole load calculation and leads to equipment that is sized for a climate the building is not actually in. Real building data must be paired with the correct local design temperature and humidity.

#131 • Installation Planning

What building data must be gathered before a commercial load calculation can be performed?

Answer

Floor area, wall and roof construction and insulation values, window area and orientation, occupancy schedule, internal equipment heat gain, and local design temperature and humidity conditions for the specific city. Commercial and light commercial calculations are typically done using recognized methods built on ASHRAE fundamentals. Missing or wrong data at this stage produces an inaccurate load regardless of how careful the arithmetic is.

#132 • Installation Planning

What is sensible heat ratio (SHR) and why does a mismatch between equipment SHR and building SHR cause problems?

Answer

SHR is the proportion of sensible heat to total heat that a system is designed to remove. If the selected equipment's SHR does not match the actual building load's SHR, the system will short cycle on the thermostat before it has run long enough to remove enough moisture. This is common in oversized residential AC installations and leaves the space feeling cold but clammy.

#133 • Installation Planning

Why can a gym have a very high latent load even with only a modest sensible load?

Answer

Occupant respiration and perspiration are latent heat sources, and a space with a lot of occupants generates a large amount of moisture regardless of how much sensible heat the space itself produces. The gym's sensible load might come mainly from lighting and modest solar gain, but the crowd of exercising occupants drives the latent load much higher. A load calculation that only tracks total kW without separating the two would miss this.

#134 • Installation Planning

One ton of refrigeration equals approximately how many kilowatts?

Answer

One ton of refrigeration equals approximately 3.52 kW. Metric-rated equipment specification sheets list capacity directly in kW rather than tons. A mechanic converting between a ton-rated spec sheet and a kW-based load calculation needs this conversion to compare them correctly.

#135 • Installation Planning

Explain two ways oversizing a refrigeration or AC compressor damages the equipment over its service life.

Answer

An oversized unit satisfies the thermostat quickly, so it experiences far more start cycles than a properly sized unit, and every start causes higher electrical inrush and more wear on contacts, bearings and motor windings. Over years this shortens compressor life significantly and drives up electricity costs from repeated starting current draw. The short run time also means the system never runs long enough to properly dehumidify the space.

#136 • Installation Planning

What operational problems does an undersized refrigeration or AC system cause?

Answer

An undersized system runs continuously and still cannot reach setpoint on the hottest or most loaded days, leading to occupant complaints, product spoilage in refrigeration, or process failures industrially. Continuous run time at maximum capacity also leaves no headroom for equipment degradation over time. A system already at its design limit on day one falls further behind as coils foul and charge drifts.

#137 • Installation Planning

What sizing margin should a mechanic target when matching equipment capacity to a calculated load?

Answer

Equipment rated capacity at the actual design conditions the site will experience should closely match the calculated load, generally within about 10 percent. The mechanic should size for a small safety margin rather than a large one. Oversizing beyond that margin causes short cycling and premature compressor wear.

#138 • Installation Planning

Why can't a mechanic read a compressor's rated capacity off the spec sheet and apply it to every job condition?

Answer

A compressor's rated capacity on the spec sheet is only valid at the stated rating conditions, such as a specific outdoor ambient, entering air or water temperature, and evaporating temperature. Manufacturer capacity tables must be read at the correct entering conditions for the actual site, not assumed to hold everywhere. Applying a table value from the wrong condition can lead to an under or oversized selection.

#139 • Installation Planning

A new AC unit oversized for its space satisfies the thermostat fast but leaves rooms feeling clammy. Name the two mechanisms at work.

Answer

First, the mismatch between the equipment's sensible heat ratio and the building's actual SHR means moisture is not removed before the thermostat is satisfied. Second, the short run time itself cuts off dehumidification before the coil has been running long enough to condense meaningful moisture. Both problems trace back to selecting capacity without a proper load calculation.

#140 • Installation Planning

Describe the design of a reciprocating compressor and the application it suits best.

Answer

Reciprocating compressors use pistons in cylinders and suit a wide range of capacities and applications, particularly where high compression ratios are needed, such as low temperature commercial refrigeration. They tolerate a wide operating envelope and are relatively easy to service and rebuild in the field. This makes them common on older commercial racks and many industrial systems still in service.

#141 • Installation Planning

How does a scroll compressor compress refrigerant, and why has it become the standard for residential air conditioning?

Answer

Scroll compressors use two interleaving spiral scrolls, one fixed and one orbiting, to compress refrigerant in a smooth, largely continuous process. They have fewer moving parts than reciprocating compressors, run quieter, and tolerate small amounts of liquid slugging better. This reliability and efficiency at typical air conditioning compression ratios made them the standard choice for residential and light commercial systems.

#142 • Installation Planning

Why are screw compressors the standard choice for ammonia industrial refrigeration plants?

Answer

Screw compressors are rotary twin screw designs used where large capacities are needed, providing smooth, continuous compression. Equipped with capacity slide valves for modulation, they handle a wide range of loads efficiently, which suits industrial ammonia plants and large chiller applications. They are the standard for these large centrifugal-alternative chiller and industrial ammonia systems.

#143 • Installation Planning

What compressor type is typically used in window air conditioners and small dehumidifiers, and what is its main limitation?

Answer

Rotary compressors, including rolling piston and rotary vane designs, are used mainly in small capacity applications such as window air conditioners, small dehumidifiers, and some small commercial equipment. They are compact and low cost for their capacity range. Their main limitation is that they are generally restricted to lower capacity and lower compression ratio applications than reciprocating, scroll, or screw compressors.

#144 • Installation Planning

Compare reciprocating and scroll compressors on tolerance to liquid slugging.

Answer

Scroll compressors tolerate small amounts of liquid slugging better than reciprocating compressors do. Reciprocating compressors use pistons in cylinders that are more vulnerable to damage from liquid entering the cylinder. This is one reason scroll compressors became the standard choice for residential and light commercial equipment.

#145 • Installation Planning

A design engineer needs a compressor for a large industrial ammonia refrigeration plant. Which compressor type fits and why?

Answer

A screw compressor fits because these rotary twin screw designs are used where large capacities are needed, especially in industrial refrigeration and large chiller applications. They provide smooth, continuous compression and, equipped with capacity slide valves, handle a wide range of loads efficiently. They are the standard choice for ammonia industrial refrigeration plants.

#146 • Installation Planning

A mechanic needs a compressor for a low temperature commercial refrigeration application requiring a high compression ratio. Which type fits best?

Answer

A reciprocating compressor fits best because this design is well suited to a wide range of capacities and applications, particularly where high compression ratios are needed, such as low temperature commercial refrigeration. It tolerates a wide operating envelope. Reciprocating units are also relatively easy to service and rebuild in the field, which supports ongoing maintenance of low temperature systems.

#147 • Installation Planning

What factors should guide selecting a compressor type for a given application?

Answer

The application's capacity range, compression ratio, refrigerant, duty cycle, and serviceability requirements should all be matched to the compressor's design strengths. Selecting a compressor type means genuinely weighing these factors, not defaulting to whatever the mechanic is most familiar with installing. Reciprocating, scroll, screw, and rotary compressors each have distinct strengths across these variables.

#148 • Installation Planning

How does a screw compressor achieve capacity modulation to match varying loads?

Answer

Screw compressors are equipped with capacity slide valves that allow them to handle a wide range of loads efficiently. This lets one screw compressor modulate output rather than only running at full capacity or cycling off entirely. This modulation ability is part of why screw compressors are the standard for large industrial and chiller applications with varying loads.

#149 • Installation Planning

Name two operational advantages a scroll compressor has over a reciprocating compressor.

Answer

Scroll compressors have fewer moving parts than reciprocating compressors and run quieter. They also tolerate small amounts of liquid slugging better than reciprocating designs. These advantages, combined with reliability and efficiency at typical air conditioning compression ratios, made scroll the standard for residential and light commercial equipment.

#150 • Installation Planning

Describe an air-cooled condenser and its main efficiency limitation.

Answer

Air-cooled condensers reject heat directly to outdoor ambient air using a fan-driven coil, requiring no water supply and having low maintenance requirements. This makes them the default choice for most residential and light commercial systems. Their main limitation is that condensing temperature rises with ambient air temperature, reducing efficiency on the hottest days, exactly when system load is highest.

#151 • Installation Planning

What safety concern is specifically associated with water-cooled condensers using an open cooling tower loop?

Answer

Water treatment is required to control scale and biological growth, including legionella control, which is a genuine safety concern with any open cooling tower system. Water-cooled condensers also need freeze protection in winter and add complexity and cost compared to air-cooled units. In exchange, they can maintain a lower, more stable condensing temperature since water temperature varies less than air temperature.

#152 • Installation Planning

How does an evaporative condenser achieve a lower condensing temperature than an air-cooled condenser?

Answer

Refrigerant runs through a coil that is sprayed with water while air is drawn across it, using evaporative cooling to reach a condensing temperature close to the outdoor wet bulb temperature rather than the higher dry bulb temperature an air-cooled unit sees. Wet bulb temperature is generally lower than dry bulb, so this gives much of the efficiency benefit of water cooling. It does this with a self-contained water system rather than a full cooling tower loop.

#153 • Installation Planning

Compare condensing temperature stability between water-cooled and air-cooled condensers.

Answer

Water-cooled condensers can maintain a lower, more stable condensing temperature than an air-cooled unit because water temperature varies less than outdoor air temperature. This improves compressor efficiency and allows a smaller compressor to do the same work. Air-cooled condensing temperature instead rises directly with ambient air temperature, cutting efficiency on the hottest days.

#154 • Installation Planning

A large commercial ammonia system needs good efficiency but a full cooling tower loop is not justified. Which condenser type fits?

Answer

An evaporative condenser fits well here, since it is common on larger commercial and industrial ammonia and CO₂ systems where efficiency matters but a full cooling tower loop is not justified. It achieves a condensing temperature close to the outdoor wet bulb temperature using a self-contained sprayed coil. This gives much of the efficiency benefit of water cooling without the added water loop complexity.

#155 • Installation Planning

What factors does a mechanic weigh when helping select or confirm condenser type at the design stage?

Answer

Available water supply and treatment capability, noise limits, available space, winter climate, and required efficiency are all weighed when selecting condenser type. Air-cooled, water-cooled, and evaporative condensers each trade off differently on these factors. The right choice depends on the specific site conditions rather than one condenser type being universally best.

#156 • Installation Planning

Why does an air-cooled condenser lose efficiency at exactly the time system load is highest?

Answer

Condensing temperature on an air-cooled unit rises directly with outdoor ambient air temperature, since heat is rejected straight to that air. On the hottest days ambient temperature is highest, so condensing temperature and compressor work both rise while system cooling load is also at its peak. This is the main limitation of air-cooled condensers compared to water-cooled or evaporative designs.

#157 • Installation Planning

List two tradeoffs a designer accepts by choosing a water-cooled condenser over an air-cooled one.

Answer

A water-cooled condenser adds the complexity, cost and maintenance of a water loop, including water treatment for scale and biological growth control such as legionella. It also requires freeze protection in winter, unlike an air-cooled unit rejecting directly to air. In exchange it gains a lower, more stable condensing temperature and improved compressor efficiency.

#158 • Installation Planning

What two competing needs must refrigerant piping design balance?

Answer

Piping must keep pressure drop low enough that system capacity and efficiency are not wasted, while keeping refrigerant velocity high enough, particularly in suction lines, to carry oil back to the compressor. These two needs pull in opposite directions since a larger pipe lowers pressure drop but also lowers velocity. Correct sizing finds the balance point the equipment manufacturer's limits allow.

#159 • Installation Planning

How does pressure drop in a suction line reduce compressor capacity?

Answer

Every kPa of pressure drop in the suction line effectively lowers the saturated suction temperature the compressor sees, which reduces compressor capacity and efficiency. Piping is sized generously enough, and kept short and direct enough, to keep these losses within the limits the equipment manufacturer specifies. Excess pressure drop represents lost system capacity even though nothing is physically broken.

#160 • Installation Planning

What happens if liquid line pressure drop is large enough to bring refrigerant below its saturation pressure?

Answer

Flash gas forms in the liquid line before the refrigerant reaches the expansion device. This starves the evaporator of usable refrigerant and causes control problems at the expansion valve. Liquid piping is sized generously enough and kept short and direct enough to avoid this.

#161 • Installation Planning

How is oil return managed in a horizontal suction line versus a vertical suction riser?

Answer

In horizontal suction lines oil return is managed mainly through correct slope, typically toward the compressor. In vertical suction risers gravity works against oil return, so the pipe must instead be sized small enough that refrigerant vapour velocity stays high enough to physically carry oil droplets upward with the flow. These are two different mechanisms for the same underlying problem of getting oil back to the compressor.

#162 • Installation Planning

What refrigerant vapour velocity is generally needed in a vertical suction riser to carry oil upward?

Answer

Vertical suction risers generally need refrigerant vapour velocity of at least about 5 to 8 metres per second to physically carry oil droplets upward with the vapour flow. Below that velocity, oil falls back down the pipe rather than being carried along. This is why riser sizing is driven by velocity, not just by minimizing pressure drop.

#163 • Installation Planning

A system was piped for future capacity that never got added, and the suction riser is now oversized for actual load. What failure mode results?

Answer

Oil will fall back down the pipe and pool in the riser or in the compressor crankcase over time, because the riser is too large for the actual refrigerant velocity to carry oil upward. This eventually leads to a compressor lubrication failure. The riser should have been sized to the actual current load's velocity, not a hoped-for future load.

#164 • Installation Planning

What design technique allows one suction riser to handle both low load and high load conditions without losing oil return?

Answer

Double suction risers with a trap are a standard technique used where load, and therefore refrigerant velocity, is expected to vary widely. One riser is sized for oil return at low load, while a second riser only carries flow at high load. This lets the system maintain adequate oil-carrying velocity across the full range of operating loads.

#165 • Installation Planning

Why does oversizing a vertical suction riser cause a compressor lubrication problem rather than just a minor efficiency loss?

Answer

An oversized riser reduces refrigerant vapour velocity below the roughly 5 to 8 metres per second needed to carry oil droplets upward against gravity. Oil then falls back and pools in the riser or the crankcase instead of returning continuously to the compressor. Over time this starves the compressor of lubrication and causes a lubrication failure, a mechanical problem rather than just lost capacity.

#166 • Installation Planning

On a P&ID;, what do circle symbols generally represent?

Answer

Circles on a P&ID; represent instruments and gauges. Instrument tags on the drawing identify what each device measures, such as a pressure transmitter or temperature sensor, and where its signal goes. Reading these symbols fluently lets a mechanic understand system operation and safety devices from the drawing alone.

#167 • Installation Planning

What information do P&ID; valve symbols convey to a mechanic reading a drawing?

Answer

Valve symbols indicate the valve type, such as ball, globe, check, solenoid, or relief. Combined with specific line weights and styles for different pipe types, they let a mechanic identify what kind of valve sits at each point in the system without physically tracing every line in the field. This is especially valuable on larger industrial systems with dozens of valves and control points.

#168 • Installation Planning

What does a mechanical drawing's equipment schedule or legend define, and why does the mechanic check it against delivered equipment?

Answer

The schedule or legend defines equipment tags, sizes, and capacities shown on the plan and section views. The mechanic cross references the equipment schedule against the physical nameplate data on delivered equipment to confirm the correct unit was supplied before installation begins. Skipping this check risks installing the wrong equipment and only discovering it after work has started.

#169 • Installation Planning

Give an example of a costly error that accurate drawing reading prevents before work starts.

Answer

Running a liquid line where a suction line was intended, or missing a required isolation valve that the safety and service plan depends on, are both costly errors accurate drawing reading prevents. Mechanical drawings show equipment locations, piping routing, and clearances in plan and section view specifically so these details are confirmed before pipe is cut. Catching an error on paper is far cheaper than correcting installed piping.

#170 • Installation Planning

Why is fluent P&ID; symbol reading especially valuable on large industrial systems?

Answer

Large industrial systems can have dozens of valves and control points, and standardized symbols for instruments, valve types, and pipe types let a mechanic understand system operation and safety devices from the drawing alone. This avoids having to physically trace every line in the field to figure out what a component does. Instrument tags also show where each device's signal goes into the control system.

#171 • Installation Planning

What is CSA B52 and what does it govern?

Answer

CSA B52, the Canadian Mechanical Refrigeration Code, is the governing code for the design, construction, installation and operation of refrigeration systems across Canada. It sets requirements for refrigerant classification and allowable quantities by occupancy type, machine room construction and ventilation, required safety devices, piping material and pressure ratings, and labelling and signage. A journeyperson mechanic is expected to apply these requirements when planning and installing a system, not simply follow a drawing.

#172 • Installation Planning

Name three categories of requirement that CSA B52 sets for a refrigeration installation.

Answer

CSA B52 sets requirements for refrigerant classification and allowable quantities by occupancy type, machine room construction and ventilation, and required safety devices such as pressure relief valves and high pressure cutouts. It also covers piping material and pressure ratings and labelling and signage requirements. Together these define how a system must be designed and built across Canada.

#173 • Installation Planning

What must typically be submitted to the authority having jurisdiction as part of the permit process for a new refrigeration installation?

Answer

The permit process typically requires submission of design drawings, equipment specifications, and refrigerant charge calculations. This lets the inspecting authority confirm the proposed installation meets CSA B52 and the applicable provincial gas and refrigeration regulations. New installations above certain size and refrigerant quantity thresholds require this permit before work begins.

#174 • Installation Planning

What are the consequences of working without a required permit or proceeding past a failed inspection point uncorrected?

Answer

Working without a required permit, or proceeding past a failed inspection point without correcting the deficiency, exposes both the mechanic and the employer to liability. It can also invalidate insurance coverage on the building. Completed installations are subject to inspection before they can legally be put into service.

#175 • Installation Planning

When does a new refrigeration or air conditioning installation require a permit before work begins?

Answer

New installations above certain size and refrigerant quantity thresholds require a permit from the local authority having jurisdiction before work begins. The completed installation is then subject to inspection before it can legally be put into service. The exact thresholds are set by the applicable provincial gas and refrigeration regulations alongside CSA B52.

#176 • Installation Planning

Why is a journeyperson mechanic expected to understand the code basis behind a drawing, not just follow it?

Answer

A journeyperson mechanic is expected to know how to apply CSA B52 requirements when planning and installing a system, not simply follow a drawing without understanding the code basis behind it. A drawing can contain an error or omission, and a mechanic who understands the underlying code can catch it. Blindly following a drawing offers no protection against a design mistake that violates code.

#177 • Installation Planning

What must a permit application's refrigerant charge calculation be checked against?

Answer

The calculated charge is checked against the maximum allowable quantity for the room's classification and ventilation rate under CSA B52. Getting this estimate close before construction avoids a costly redesign if the calculated permit-stage charge turns out to exceed what the room's ventilation can safely handle. This is part of why charge is estimated early at the design stage, not left until commissioning.

#178 • Installation Planning

Why must a mechanic confirm structural support based on operating weight rather than shipping weight?

Answer

Operating weight includes the equipment with its refrigerant and oil charge, which is heavier than the empty shipping weight. Structural support has to be confirmed adequate for this actual operating weight, and roof mounted equipment needs an engineered curb or support structure sized for the actual unit. Using shipping weight would understate the real load the structure must carry long term.

#179 • Installation Planning

Why does recirculation of hot discharge air back into an air-cooled condenser's intake matter for clearance planning?

Answer

Recirculation of hot discharge air back into the intake degrades condenser performance, since the condenser then draws in air already warmed by its own discharge instead of cooler ambient air. Required clearances around the equipment for airflow, particularly around air-cooled condensers, must be confirmed against the manufacturer's installation instructions and against CSA B52 clearance requirements. Adequate clearance keeps discharge air from being pulled back into the intake.

#180 • Installation Planning

What is the purpose of machine room ventilation requirements under CSA B52?

Answer

Mechanical or machine rooms housing refrigeration equipment have ventilation requirements sized to the refrigerant type and quantity present. This ensures that in the event of a leak the room can be purged to a safe concentration, with detection and alarm systems triggering mechanical ventilation before concentrations reach a hazardous level. This ventilation requirement is directly tied to the refrigerant's safety group classification.

#181 • Installation Planning

Under ASHRAE Standard 34, what does the toxicity axis (A versus B) of a refrigerant safety group indicate?

Answer

ASHRAE Standard 34, referenced by CSA B52, classifies refrigerants into safety groups based on toxicity, with A meaning lower toxicity and B meaning higher toxicity. This is combined with a separate flammability rating to form the full safety group, such as A1 or A2L. The toxicity rating affects placement restrictions alongside the flammability rating.

#182 • Installation Planning

Under ASHRAE Standard 34, what do the flammability numbers 1, 2L, 2 and 3 mean?

Answer

The flammability axis rates refrigerants from 1 for no flame propagation, through 2L for lower flammability and 2 for flammability, up to 3 for higher flammability. This is combined with the A or B toxicity rating to form the full safety group classification, such as A1 or A2L. Both axes together determine placement restrictions under CSA B52.

#183 • Installation Planning

Give an example of an A1 refrigerant and describe its placement restrictions.

Answer

R-410A and R-134a are examples of A1 refrigerants, meaning non-flammable and low toxicity. An A1 refrigerant carries the fewest placement restrictions of any safety group. This makes A1 refrigerants generally the simplest to site from a code perspective.

#184 • Installation Planning

Give an example of an A2L refrigerant and describe why the industry has moved toward it.

Answer

R-32 and R-454B are examples of A2L refrigerants, both increasingly common as the industry moves toward lower global warming potential refrigerants. A2L means mildly flammable and lower toxicity. Systems using A2L refrigerants carry additional requirements compared to A1 refrigerants because of that mild flammability.

#185 • Installation Planning

What additional design requirements apply to systems using an A2L refrigerant compared to an A1 refrigerant?

Answer

Systems using A2L refrigerants have additional requirements around allowable charge size in occupied spaces and ignition source separation. In some cases mechanical ventilation or refrigerant detection is also required, depending on the application and charge amount. These requirements exist because A2L refrigerants are mildly flammable, unlike non-flammable A1 refrigerants.

#186 • Installation Planning

A mechanic is planning equipment placement for a system using R-454B in an occupied space. What must be confirmed first?

Answer

The refrigerant's ASHRAE 34 safety group classification, A2L in this case, must be confirmed at the design stage before equipment location is finalized. This determines allowable charge size in the occupied space, ignition source separation, and possibly mechanical ventilation or detection requirements. Knowing the safety group is not optional trivia, it directly determines where the equipment legally can and cannot be installed.

#187 • Installation Planning

What information is used to estimate refrigerant charge at the design stage?

Answer

Refrigerant charge is estimated using piping length, pipe diameter, and equipment internal volume. This lets the correct refrigerant quantity be ordered and lets the calculated charge be checked against the maximum allowable quantity for the room's classification and ventilation rate under CSA B52. The estimate is done at the design stage, before construction, not left until commissioning.

#188 • Installation Planning

Why is it important to estimate refrigerant charge accurately before construction rather than waiting until commissioning?

Answer

Getting the charge estimate close before construction avoids a costly redesign if the calculated permit-stage charge turns out to exceed what the room's ventilation can safely handle under CSA B52. Charge is estimated from piping length, pipe diameter, and equipment internal volume during design. Finding out at commissioning that the charge exceeds the room's allowable quantity would require reworking piping or ventilation after installation.

#189 • Installation Planning

What is heat recovery in a refrigeration or AC system, and what can the captured heat be used for?

Answer

Heat recovery systems capture rejected condenser heat and use it productively instead of discharging it to atmosphere. It can be used for domestic hot water preheat, space heating, or process heat. This can substantially reduce a building's overall energy use where a simultaneous heating and cooling demand exists.

#190 • Installation Planning

What do variable frequency drives (VFDs) modulate in a refrigeration or AC system, and why does this improve efficiency?

Answer

VFDs allow compressor motors, condenser fans, and pump motors to modulate speed to match actual load rather than cycling on and off at fixed speed. This improves part load efficiency significantly, since most systems spend the majority of their operating hours at partial load, not full design load. Matching motor speed to actual demand avoids the energy waste of fixed-speed cycling.

#191 • Installation Planning

Why does VFD-based part load efficiency matter so much if a system is only rarely at full design load?

Answer

Most systems spend the majority of their operating hours at partial load, not full design load, so efficiency at partial load has a larger effect on total energy use than efficiency at the rare peak-load condition. VFDs allow compressor motors, condenser fans, and pump motors to modulate speed to match actual load. This makes VFD-driven part load efficiency a major lever for reducing overall building energy use.

#192 • Installation Planning

Compare staged capacity control and fully modulating capacity control on load matching and cost.

Answer

Staged control uses multiple compressors or multiple stages within one compressor, while modulating control uses a VFD or unloading mechanism to vary output continuously. Modulating control generally achieves closer load matching and smoother operation than staged control alone, but at a higher initial equipment cost. Selecting between them weighs capital cost against expected energy savings and control precision over the system's service life.

#193 • Installation Planning

A designer is choosing between staged and fully modulating capacity control for a new system. What tradeoff are they weighing?

Answer

They are weighing capital cost against expected energy savings and control precision over the system's service life. Modulating control, using a VFD or unloading mechanism, generally achieves closer load matching and smoother operation than staged control, but at a higher initial equipment cost. Staged control, using multiple compressors or stages, is less precise but cheaper to install.

#194 • Installation Planning

A building has a simultaneous heating and cooling demand. What energy efficiency design feature is well suited to this situation?

Answer

Heat recovery is well suited here, since it captures rejected condenser heat and uses it productively for domestic hot water preheat, space heating, or process heat instead of discharging it to atmosphere. It can substantially reduce overall building energy use specifically where a simultaneous heating and cooling demand exists. Without that simultaneous demand, there is less rejected heat available to usefully recover at the times it is needed.

#195 • Installation Planning

Why is energy efficiency now treated as a core design consideration rather than an afterthought in system planning?

Answer

Tools like heat recovery, VFDs, and capacity control strategy all directly affect a building's energy use and operating cost over the system's service life, not just its ability to meet peak load. Heat recovery can substantially cut energy use where simultaneous heating and cooling demand exists, and VFDs improve part load efficiency across the majority of operating hours most systems actually run at. Building these considerations into the design stage, alongside load calculation and equipment sizing, is what produces a system that is efficient to operate, not just adequate to meet peak demand.

#196 • Installation

What is the key difference in how ACR tube and Type L copper are sized?

Answer

ACR tube is sized by actual outside diameter (OD) and is the standard for field-run refrigerant lines, sold cleaned, capped, and dehydrated from the factory. Type L copper is sized by nominal inside diameter and is more common in plumbing and hydronic work. A mechanic must never assume Type L and ACR of the same nominal size are interchangeable without checking actual OD.

#197 • Installation

Where is Type K copper tubing typically used, and why is it uncommon on standard refrigerant circuits?

Answer

Type K has the thickest wall of the three copper designations and appears mainly in underground or high-pressure applications. Its extra wall thickness is not needed for standard refrigerant circuits, so it is rarely used there. ACR tube remains the standard choice for field-run refrigerant lines instead.

#198 • Installation

Why is ACR tube preferred for field-run refrigerant lines beyond its OD-based sizing?

Answer

ACR tube is sold cleaned, capped, and dehydrated straight from the factory, which keeps moisture and contaminants out of the tube until it is installed. Type L copper is intended for plumbing and hydronic work and does not come with this refrigeration-specific factory preparation. This makes ACR the correct choice even though tubing sizes are still specified in inch fractions on the job.

#199 · Installation

Why must copper tubing be cut with a proper tubing cutter instead of a hacksaw?

Answer

A hacksaw introduces metal filings into the tube that a rotary cutter wheel avoids entirely. Those filings can travel through the system and damage components. A tubing cutter also produces a cleaner, more consistently round cut for proper fitting seating.

#200 · Installation

A technician advances a tubing cutter with heavy pressure in one pass to save time. What is wrong with this?

Answer

The cutter wheel should be advanced gradually, a quarter turn at a time, rotating the cutter fully around the tube between advances. Rushing the cut with heavy pressure produces a pinched, oval end instead of a round one. An out-of-round cut will not seat properly in a fitting and can leak.

#201 • Installation

Why must a tube end be deburred after cutting, and how should the mechanic hold the tube while reaming?

Answer

An un-reamed tube restricts flow at the joint, and the burr can shear off during operation and travel downstream to plug a metering device or valve seat. Deburring must remove the burr fully while keeping the tube round. The mechanic should tip the tube downward while reaming so shavings fall out rather than into the tube.

#202 • Installation

Why is brazing the primary method for permanently joining refrigerant piping?

Answer

A properly brazed joint is stronger than the tube itself and, unlike a mechanical fitting, has no gasket or seal that can degrade over time. The filler metal fuses tube and fitting into a continuous piece of metal with effectively no leak path. Its main weakness is that it cannot be reasonably disassembled without cutting the joint out.

#203 · Installation

Why does the fit-up gap between tube and fitting socket matter for a brazed joint?

Answer

A brazed joint relies on capillary action drawing molten filler metal into the small clearance between the tube and the fitting socket. Too loose a fit and capillary action fails, leaving voids in the joint. Too tight a fit and the filler cannot penetrate the joint at all.

#204 · Installation

Why should mating surfaces before brazing be abraded with emery cloth rather than steel wool?

Answer

Steel wool leaves fine particles that contaminate the copper and can initiate corrosion. Emery cloth or sandpaper cleans the surface without leaving that contamination behind. Clean, dry surfaces are essential because any oxide, oil, or moisture prevents the filler from wetting the base metal properly.

#205 • Installation

Why is flux not required when brazing a copper-to-copper joint with phos-copper filler?

Answer

Phosphorus-bearing filler alloy, commonly called Phos-Copper, acts as a self-fluxing agent on copper because its phosphorus content shields the surface during heating. Flux is only necessary when brass or steel is part of the joint. Those base metals get no benefit from the phosphorus self-fluxing effect and will oxidize rapidly under heat without a flux coating.

#206 • Installation

A mechanic is brazing a copper line to a brass service valve. Does this joint need flux?

Answer

Yes, flux is required whenever brass or steel is part of the joint, such as copper-to-brass service valves or copper-to-steel adapters. These base metals do not benefit from the phosphorus self-fluxing effect that copper-to-copper joints get from Phos-Copper filler. Flux should be applied sparingly with a brush, only to the area to be heated, to avoid contaminating the system.

#207 · Installation

During brazing, where should the torch heat be directed first, and where should the filler rod be introduced?

Answer

Heat is directed first at the base of the fitting socket and moved around the joint to bring both tube and fitting to a uniform temperature. The filler rod is then introduced at the joint mouth, not directly into the flame. The filler should flow into the joint by capillary action and form a smooth, continuous fillet around the full circumference.

#208 · Installation

What internal problem does nitrogen purging during brazing prevent, and how does it work?

Answer

When copper is heated with oxygen and moisture trapped inside the pipe, a black internal oxide scale forms on the inner wall, which later breaks loose and can plug filter driers, score compressor bearings, and contaminate the oil. Dry nitrogen fed through the piping at a low flow rate displaces the oxygen inside the tube so no internal oxidation can occur. Flow should continue before, during, and after brazing until the joint has cooled enough that oxidation is no longer possible.

#209 · Installation

Where are flare joints used instead of brazed joints, and why?

Answer

Flares are used almost exclusively where a joint must be serviceable, meaning it may need to be disconnected in the field. Examples include service valves on compressors and condensing units, some sight glasses, and line-set connections for ductless split systems. These locations need connections made and broken routinely during commissioning or service.

#210 · Installation

A mechanic flares a tube end and then realizes the flare nut is still in the toolbox. What went wrong?

Answer

The flare nut must be slid onto the tube before flaring begins. This is a step that is easy to forget, and forgetting it forces the mechanic to redo the flare since the nut cannot be added afterward. Producing a good flare also starts with a square, deburred cut.

#211 · Installation

Why does the tube stick-up in a flaring block matter when producing a flare?

Answer

The tube is clamped in the flaring block with the stick-up specified by the tool manufacturer. Too much stick-up produces an oversized flare that cracks, and too little stick-up produces an undersized flare that will not seat. Getting the stick-up right is essential to a leak-free, serviceable joint.

#212 · Installation

What should a mechanic check before assembling and torquing a completed flare joint?

Answer

The flare should be checked visually for a uniform cone with no cracks before the joint is assembled and torqued to specification. Overtightening the flare nut can also crack the flare, so torque should be applied to specification rather than by feel. A cracked or malformed flare will leak under pressure.

#213 · Installation

What is swaging, and what advantage does it offer over a standard coupling fitting?

Answer

Swaging expands the end of one tube so a second tube of the same nominal size can be inserted directly into it without a coupling fitting, and the joint is then brazed. This produces a joint with fewer components and one less potential leak point than a standard coupling. A swaging tool expands the tube in stages to avoid splitting the copper.

#214 · Installation

How deep should a swaged tube end be expanded relative to the inserted tube's OD?

Answer

The swaged depth should equal or slightly exceed the outside diameter of the tube being inserted. This gives adequate surface area for the braze to form a strong, capillary-fed joint. A swaging tool expands the tube gradually in stages to avoid splitting the copper.

#215 · Installation

Why does sagging refrigerant piping between supports cause a problem beyond appearance?

Answer

Sagging creates low points where oil pools and unwanted traps form along the run. This disrupts intended oil return and can starve other parts of the system of lubrication. Supports must be spaced closely enough to prevent this sagging while still letting the pipe move without stress concentrating at any one point.

#216 · Installation

How does support spacing differ between small flexible tubing and larger rigid pipe?

Answer

Horizontal runs are supported at regular intervals, tighter for smaller, more flexible tubing and wider for larger, rigid pipe. Additional support is added near every change of direction and near heavy in-line components such as filter driers or solenoid valves. This prevents the weight of those components from hanging on the tubing alone.

#217 · Installation

What two things absorb compressor vibration so it does not fatigue the connected rigid piping?

Answer

Compressors are typically mounted on rubber-in-shear or spring isolators at the base. The lines connecting to the compressor should include a flexible connector or vibration eliminator hose, plus at least one 90 degree offset or loop in the rigid piping nearby. Together these absorb cyclic vibration rather than carrying it straight down a stiff run.

#218 · Installation

A pipe clamp is installed rigidly very close to the compressor discharge line. Why does this defeat the vibration isolation?

Answer

Rigid clamps installed too close to the compressor defeat the isolators and flexible hose entirely, because vibration simply finds another rigid path through the clamp. The flexible connector and offset loop only work if the piping near the compressor is otherwise free to flex. This can fatigue the pipe and transmit noise through the building structure over time.

#219 · Installation

Why is a sleeve used where refrigerant piping passes through a wall, floor, or roof?

Answer

A sleeve keeps the pipe from being clamped directly by the structure at the penetration. This allows the necessary thermal movement of the pipe as it expands and contracts with temperature. It also keeps vibration from transferring into the building structure.

#220 · Installation

Why do vertical suction risers pose a greater oil return challenge than horizontal suction lines?

Answer

On horizontal suction lines, gas velocity alone is usually enough to keep oil moving along with the refrigerant. On vertical risers, gravity constantly pulls oil back down the pipe wall, so refrigerant velocity has to be high enough to carry it upward against gravity. If design velocity is too low, oil accumulates at the base of the riser instead of returning to the compressor.

#221 · Installation

What happens if oil accumulates at the base of a vertical suction riser due to low velocity?

Answer

The oil eventually forms a slug that can hit the compressor and cause serious mechanical damage. In the meantime, the compressor is starved of the lubrication it needs since the oil is not returning with the refrigerant. This is why riser velocity design is critical, not just a comfort issue.

#222 · Installation

What is the function of a trap placed at the base of a vertical suction riser?

Answer

The trap collects any oil that has fallen out of suspension in a small pocket at the base of the riser. As gas continues to move past, it periodically picks that oil back up and carries it upward. This prevents the oil from simply pooling indefinitely at the base of the riser.

#223 · Installation

How often should additional traps be placed on a tall vertical suction riser?

Answer

Additional traps are placed at intervals up a tall riser, generally every 6 to 8 metres or per the manufacturer's engineering data. Re-entrained oil can fall out of suspension again over a long run. Periodic traps keep picking that oil back up rather than letting it pool over a long vertical distance.

#224 · Installation

What oil return problem does capacity modulation create for a single fixed-size suction riser?

Answer

A riser sized for full-load velocity will be oversized at minimum load, and velocity at low capacity may drop below the minimum needed to carry oil upward at all. This happens when a compressor runs at reduced capacity through unloading or staging. A single riser cannot maintain adequate oil-carrying velocity across the full range of capacity.

#225 · Installation

Compare how gas flows through a double riser at minimum load versus at increased load.

Answer

At low load, all gas is forced through the small riser at high velocity, which keeps oil moving properly despite the low flow. As load and gas volume increase, a trap arrangement between the two risers lets gas flow through the larger pipe as well, once there is enough volume to maintain adequate velocity in both. This keeps oil return effective across the full range of capacity rather than at one design point alone.

#226 · Installation

Compare the leak path and strength of a properly made brazed joint versus a mechanical flare or compression fitting.

Answer

A properly made brazed joint is permanent and extremely strong, with effectively no leak path since the filler fuses tube and fitting into a continuous piece of metal. A flare or compression fitting relies on a mechanical seal that can loosen from vibration, thermal cycling, or age. Brazed joints cannot be reasonably disassembled, while mechanical fittings can be disconnected and reconnected without heat.

#227 · Installation

Name two situations where a flare or compression fitting is the correct choice over a brazed joint.

Answer

They are correct for service valves and for connections a technician will realistically need to service equipment behind, since they can be disconnected without heat. They are also correct where hot work is not practical or permitted, such as close to combustible material or in an occupied space where a fire watch cannot reasonably be arranged. A brazed joint would require cutting out the pipe to service these connections.

#228 · Installation

What must a mechanic do at every service visit specifically because of how mechanical joints fail?

Answer

Every mechanical joint must be checked at every service visit because the mechanical seal can loosen from vibration, thermal cycling, or age. This is the tradeoff for the convenience of being able to disconnect and reconnect the joint without heat. A brazed joint does not carry this same ongoing inspection requirement since it has no seal to loosen.

#229 · Installation

What governs where mechanical joints are permitted on a refrigeration system, beyond convenience or speed?

Answer

Provincial and national codes generally restrict where mechanical joints may be used, particularly in concealed locations or high-pressure applications. A mechanic must know the code requirements for the jurisdiction and refrigerant involved. Choosing between mechanical and brazed joints is a design and code decision, not just a matter of convenience.

#230 · Installation

Name the two reasons suction lines and other cold surfaces must be insulated.

Answer

Insulation reduces heat gain into the refrigerant, which would otherwise cost system efficiency. It also prevents surface condensation on the cold pipe, which left unmanaged leads to dripping water damage and, over time, ice buildup on the pipe. Both reasons apply to any pipe surface running below the ambient dew point.

#231 · Installation

Compare closed-cell elastomeric insulation to fiberglass insulation for refrigeration piping.

Answer

Closed-cell elastomeric insulation is now the standard for refrigeration piping because its structure resists moisture absorption almost entirely. Fiberglass is open-cell and will absorb and hold moisture if its outer jacket is ever breached, which loses most of its insulating value and can accelerate corrosion of the pipe underneath. Fiberglass is now seen mainly on larger commercial ductwork rather than piping.

#232 · Installation

What happens if a vapour barrier around insulated suction line piping is breached by a gap or puncture?

Answer

Warm, moist ambient air reaches the cold pipe surface underneath the insulation, where it condenses and, on very cold suction lines, freezes. That trapped moisture cannot escape once inside the insulation, so it accumulates over time. The resulting ice can eventually split the insulation jacket open.

#233 · Installation

How should seams and butt joints in elastomeric insulation be sealed, and what is the common mistake?

Answer

All seams and butt joints must be sealed with the manufacturer's specified contact adhesive, applied to both mating surfaces and allowed to become tacky before joining. The common mistake is pressing the surfaces together dry instead of using adhesive. A continuous vapour barrier is the part of the installation most often done poorly.

#234 · Installation

At a penetration, hanger, or support, how must insulation be treated to maintain the vapour barrier?

Answer

At every penetration, hanger, support, or transition to a fitting, the insulation must be sealed back to itself or to a boot fitting. There must be no break in the vapour barrier anywhere along the run. Any single gap allows moisture to reach the cold pipe underneath.

#235 · Installation

What gas is used for pressure testing refrigeration piping, and what must never be used instead?

Answer

Pressure testing uses dry nitrogen, brought up to a standing test pressure appropriate to the system's design pressure and applicable code. Oxygen and compressed air must never be used for this purpose. The test pressure is held for a specified period, commonly hours, while the technician watches for any drop on a calibrated gauge.

#236 · Installation

During a standing pressure test, what factor besides an actual leak can cause the gauge reading to change?

Answer

Ambient temperature change affects gauge pressure independent of any leak, so the technician must account for it when reading the gauge over the test period. A confirmed drop beyond what temperature explains indicates a real leak. Only after the system holds pressure with no unexplained drop should nitrogen be released and evacuation begin.

#237 · Installation

A pressure test shows a confirmed gauge drop not explained by temperature change. What is the next step?

Answer

A confirmed drop indicates a leak that must be located, commonly with an electronic detector or soap bubble solution at every joint. The leak must be repaired before proceeding. Only after the system holds pressure with no drop should nitrogen be released and evacuation begin.

#238 · Installation

What is deep vacuum theory, and why does it justify pulling a system down to microns rather than kPa?

Answer

At very low absolute pressure, measured in microns of mercury rather than kPa or psig, the boiling point of any residual water drops low enough that it boils off as vapour and is pulled out by the vacuum pump. This is the mechanism behind evacuation removing air, nitrogen, and moisture from the system. A coarse gauge reading in kPa cannot show whether this deep a vacuum has actually been reached.

#239 · Installation

Why is a single evacuation often not sufficient to fully dry out a refrigeration system?

Answer

Moisture can be trapped in oil, in fittings, or adsorbed onto internal surfaces in a way that does not release completely on the first vacuum pass. A single evacuation may leave this trapped moisture behind even though the gauge shows a deep vacuum was reached. This is the reason the triple evacuation method is used instead.

#240 · Installation

Describe the steps of the triple evacuation method.

Answer

The system is pulled to a deep vacuum, then broken with dry nitrogen back up to a slight positive pressure, which helps release trapped moisture. This cycle repeats three times before the final deep evacuation is held. Each nitrogen break disturbs residual moisture that the previous vacuum pass could not reach directly.

#241 • Installation

Why should a micron gauge be installed away from the vacuum pump rather than right at it?

Answer

A reading taken too close to the pump can show a deeper vacuum than actually exists elsewhere in the system, due to pressure drop across long hoses or restrictive fittings. Installing the gauge away from the pump gives a reading that reflects the true vacuum level in the piping being evacuated. A standard compound gauge is not accurate enough at these pressures, so a micron gauge is required.

#242 • Installation

How is a rate-of-rise test performed after evacuation, and how does the technician interpret the result?

Answer

The technician isolates the system from the pump and observes the micron gauge over ten to fifteen minutes. A slight rise that levels off is attributable to residual moisture still evaporating and is generally acceptable. Pressure that climbs steadily without leveling off indicates a leak letting outside air back in, which must be found and repaired before evacuation and charging proceed.

#243 · Installation

How should a liquid receiver be mounted, and where does the liquid line take suction from?

Answer

Liquid receivers are typically mounted vertically with proper support for their weight when full, since they store excess refrigerant charge. They are positioned so the liquid line takes suction from the bottom of the receiver. This ensures liquid refrigerant, not vapour, is drawn out to feed the rest of the system.

#244 · Installation

What does a suction accumulator protect against, and how does trapped oil get back to the compressor?

Answer

A suction accumulator protects the compressor from liquid or oil slugging during abnormal conditions such as low load or defrost. It is piped with the correct internal oil-return orifice or U-tube so oil trapped inside meters back to the compressor rather than sitting indefinitely. Correct orientation of this internal metering feature is essential to its function.

#245 • Installation

What two things does a sight glass let a technician visually confirm?

Answer

A sight glass lets the technician visually confirm a full column of liquid, where bubbles indicate an undercharge or restriction in the system. Many sight glasses also include a moisture indicator that changes colour to warn of contamination. Together these give a quick visual check of charge and moisture condition.

#246 • Installation

Why should a filter drier be installed only after brazing is complete on that section of piping?

Answer

Installing the drier before brazing that section would let scale and flux residue from the brazing process immediately load the drier. Installing it after brazing keeps the drier's capacity available for normal system contaminants instead. The drier must also be installed with its arrow pointing in the direction of flow.

#247 · Installation

What installation requirement applies specifically to a solenoid valve's coil?

Answer

The coil must be oriented as specified by the manufacturer, typically upright. The flow arrow on the valve body must also be matched correctly to the actual system flow direction. Solenoid valves control flow electrically, so correct coil orientation is necessary for the internal mechanism to operate as designed.

#248 · Installation

What structural loads must a rooftop condensing unit's mounting pad or curb be rated to support in most of Canada?

Answer

The pad or curb, whether concrete, rubber isolation pads, or an engineered roof curb, must support the full operating weight of the unit plus wind load and, in most of Canada, snow load. Roof-mounted equipment also needs proper flashing and curb sealing to prevent water infiltration at the penetration. Enough clearance must also be left for airflow and service access.

#249 · Installation

State Ohm's Law and its two rearranged forms for finding current and resistance.

Answer

Ohm's Law is $V = I \times R$, where V is voltage, I is current, and R is resistance. Rearranged for current, $I = V / R$. Rearranged for resistance, $R = V / I$. A technician who can shuffle all three forms can diagnose most control circuit faults with just a multimeter and a schematic.

#250 · Installation

A 24V relay coil has 60 ohms of resistance. Find its power draw using $P = V^2 / R$.

Answer

$P = V^2 / R = 24^2 / 60 = 576 / 60 = 9.6W$. This is the same coil power that could also be found as $P = V \times I$ once current is known from $I = V / R$.

#251 • Installation

A 24V transformer secondary feeds a contactor coil drawing 0.3A. Find the coil's resistance and power draw.

Answer

Using $R = V / I$: $R = 24V / 0.3A = 80 \text{ ohms}$. Using $P = V \times I$: $P = 24V \times 0.3A = 7.2W$. These are the coil's normal healthy operating values.

#252 • Installation

With power off, a contactor coil that should read 80 ohms instead measures 400 ohms. What does this tell a technician?

Answer

The coil is likely partially failed and will not pull in reliably, even though it still shows some resistance instead of reading open. A healthy coil at that rated voltage and current should measure close to 80 ohms. This kind of drifted but not open reading is easy to miss if a technician checks only for continuity instead of an actual resistance value.

#253 · Installation

A compressor start winding draws 15A locked rotor through a winding resistance of 2 ohms. Find the power using $P = I^2 \times R$.

Answer

$P = I^2 \times R = 15^2 \times 2 = 225 \times 2 = 450W$. This is the heat the winding must dissipate during a locked rotor moment, which is why a protector responding to actual winding temperature matters during a prolonged locked rotor event.

#254 · Installation

A 240V defrost heater has a resistance of 9.6 ohms. Calculate the current draw using $I = V / R$.

Answer

$I = V / R = 240V / 9.6 \text{ ohms} = 25A$. This current value would be checked against the heater's nameplate rating and the circuit's overcurrent protection to confirm both are sized correctly.

#255 · Installation

Compare series and parallel circuit behavior, and explain how control circuits typically combine both.

Answer

In series, current is the same everywhere and voltage divides across each resistance, so total resistance is the sum of the parts. In parallel, voltage is the same across every branch and current divides, so total resistance is always lower than the smallest branch. Most control circuits wire switches and interlocks in series, so any one open contact stops the circuit, while loads are sometimes wired in parallel so each gets full line voltage independently.

#256 · Installation

Describe single phase power supply and how its waveform behaves over one cycle.

Answer

Single phase power is what most Canadian residences receive, typically 120V and 240V split phase, with one alternating current waveform. That waveform rises and falls to zero twice per cycle. Small residential condensers, furnace blowers, and light commercial rooftop units under a certain tonnage commonly run on single phase.

#257 · Installation

Describe three phase power and list common three phase supply voltages used in this trade in Canada.

Answer

Three phase power delivers three separate alternating waveforms, each offset by 120 electrical degrees, so the waveforms never all cross zero at the same instant. Common Canadian three phase supply voltages in this trade include 208V, 347/600V, and 575V, depending on the utility service and building type.

#258 · Installation

Why do commercial and industrial refrigeration systems typically use three phase power instead of single phase?

Answer

Because the three waveforms never all cross zero together, three phase delivers smoother, more continuous torque than single phase. This reduces mechanical stress on bearings and windings, improves starting under load, and lets a smaller conductor carry more power than single phase at the same voltage. Compressors above roughly five horsepower are typically three phase for these reasons.

#259 · Installation

A newly installed scroll compressor runs backward, draws abnormal current, and makes unusual noise. What is the likely cause?

Answer

The incoming three phase legs are connected in the wrong sequence, since scroll compressors are direction sensitive. Running backward, the compressor also fails to build pressure differential. Rotation direction should always be verified on every new three phase compressor before it is left running unattended.

#260 · Installation

How does a technician correct a three phase compressor that is rotating backward?

Answer

Swap any two of the three incoming legs, or use a phase sequence meter to confirm correct sequence before connecting. This corrects the reversed rotation caused by wrong phase sequence. Verifying rotation direction on every new three phase installation prevents leaving a backward-running compressor unattended.

#261 • Installation

Describe a permanent split capacitor (PSC) motor and where it is commonly used.

Answer

A PSC motor wires a run capacitor permanently in series with the start winding, staying in the circuit the entire time the motor runs. This gives moderate starting torque and good efficiency with few parts to fail. PSC motors are the most common single phase motor in residential condenser fans and many blowers.

#262 • Installation

Describe a capacitor-start capacitor-run (CSR) motor and why it suits compressors that start against pressure.

Answer

A CSR motor adds a start capacitor, switched in only during starting through a potential or current relay, then dropped out near 75 percent of running speed. This gives much higher starting torque than a PSC motor alone. CSR motors are common on reciprocating and older scroll compressors that must start against system pressure.

#263 · Installation

Compare PSC and CSR motors in terms of starting torque and typical application.

Answer

A PSC motor uses one run capacitor permanently in the circuit, giving moderate starting torque and fewer parts to fail. A CSR motor adds a start capacitor that switches in only during starting and drops out near 75 percent speed, giving much higher starting torque. CSR is chosen when the motor must start against system pressure, while PSC suits lighter starting loads like condenser fans.

#264 · Installation

How does a three phase induction motor produce torque without needing a capacitor?

Answer

The three phase supply itself creates a rotating magnetic field in the stator, which induces current in the rotor and produces continuous torque from start to run. No run or start capacitor is needed, unlike single phase PSC or CSR motors. This is one reason three phase motors are generally more reliable than single phase capacitor motors of similar horsepower.

#265 · Installation

What is an electronically commutated motor (ECM), and how is its speed controlled?

Answer

An ECM is a brushless DC motor controlled by an onboard module that converts incoming AC to DC and electronically switches polarity to the windings. Instead of running at one fixed speed, an ECM ramps speed smoothly to a control signal. The module, not line voltage directly, sequences energization of the windings.

#266 · Installation

Why should a technician never apply line voltage directly to an ECM's winding leads?

Answer

An ECM's onboard module must sequence energization to the windings correctly, and line voltage applied straight to the leads can destroy the motor instantly. ECMs also run around 60 percent more efficiently than PSC motors, which is a separate reason they are increasingly common. Both facts matter when diagnosing or replacing an ECM.

#267 · Installation

Compare external overload protectors and internal thermal (klixon) protectors on a compressor motor.

Answer

An external overload mounts outside the shell, often bolted to the terminals, opening the circuit when current exceeds a calibrated threshold, usually sensed via a bimetallic disc responding to current heat. An internal thermal protector is embedded inside the windings, so it responds to actual winding temperature rather than a value measured at the terminal. This makes internal protectors more accurate against a genuine overheating winding during a brief locked rotor spike or a slow overheat from poor airflow.

#268 · Installation

A compressor won't start, but continuity from common to run and common to start is normal, with none to the shell. What's the likely cause?

Answer

The windings are likely intact, so the fault is probably elsewhere, such as a protector still cooling, a failed capacitor, or a control circuit fault. Most overload protectors reset automatically once cooled, which is why a compressor tripped on internal overload may restart on its own minutes or hours later with no technician intervention. This behavior confuses many apprentices chasing an intermittent no run complaint.

#269 · Installation

What is a contactor, and what two factors determine its rating?

Answer

A contactor is an electrically operated switch that uses a low voltage control signal to control a high current or high voltage load, such as a compressor or condenser fan. Its coil is energized by the 24V control circuit, magnetically pulling contacts closed. It is rated by coil voltage, typically 24V, and by contact current rating, which must meet or exceed the running and locked rotor current of the equipment switched.

#270 · Installation

Compare a relay and a contactor, and explain why reading a relay's contact symbols matters.

Answer

A relay uses the same electromagnetic principle as a contactor but generally handles lighter loads, such as switching one low voltage signal to another or interlocking parts of a control sequence. Some relays use normally open contacts, some normally closed, and some both. Reading the schematic symbol correctly is essential to predicting what the relay actually does when its coil energizes.

#271 · Installation

What is a thermostat, and what are its two basic sensing designs?

Answer

A thermostat is a temperature sensitive switch that opens or closes a control circuit based on sensed temperature. It can be mechanical, using a bimetallic element or sealed bellows, or electronic, using a thermistor and microprocessor. Most thermostats switch the 24V control circuit rather than line voltage.

#272 · Installation

Compare what a high pressure switch and a low pressure switch each protect against.

Answer

A high pressure switch protects against excessive discharge pressure, which could rupture components or signal a condenser airflow problem. A low pressure switch protects against very low suction pressure, indicating lost charge, a restriction, or an icing evaporator. The low pressure switch also prevents compressor operation against a near vacuum, since suction gas cools the motor on many designs.

#273 · Installation

What does an anti-short-cycle timer do, and why does a compressor need one?

Answer

An anti-short-cycle timer prevents a compressor from restarting for roughly three to five minutes after it stops. This protects the compressor from starting against unequalized high side pressure. Without this delay, a quick call for cooling right after shutdown could force the compressor to start under abnormal load.

#274 · Installation

Compare fan-on delay and fan-off delay in a time delay relay's function.

Answer

A fan-on delay holds the blower off until a heat exchanger has warmed, preventing a blast of cold air at the start of a heating call. A fan-off delay keeps the blower running briefly after the heat call ends, extracting residual heat from the exchanger before shutdown. Both improve comfort and efficiency by timing fan operation to actual heat exchanger temperature.

#275 · Installation

Why must a technician read a relay's normally open versus normally closed symbol correctly before predicting circuit behavior?

Answer

A relay's contacts can be normally open, normally closed, or a mix of both on the same relay. Misreading which type a given contact is leads to a wrong prediction of what happens when the coil energizes, since a normally open contact closes on energizing while a normally closed contact opens. Correctly reading the schematic symbol is essential to accurately predicting the resulting circuit state.

#276 · Installation

How is current traced through a rung on a ladder diagram?

Answer

A ladder diagram shows two vertical power rails with individual circuits drawn as horizontal rungs between them. Reading left to right along a rung traces the current path: a power source on the left rail, through switches and interlocks in series, to a load such as a coil, and back to the other rail. This lets a technician see exactly which conditions must all be true for a load to energize.

#277 · Installation

Describe the standard ladder diagram symbols for a normally open contact, a normally closed contact, and a coil.

Answer

A normally open contact is two short parallel lines with a gap between them. A normally closed contact adds a diagonal slash through that gap. A coil is drawn as a circle or an elongated oval, and these symbols appear consistently across manufacturers with only minor variation.

#278 · Installation

A cooling contactor's rung wires the thermostat contact, both pressure switches, and a timer contact in series. What happens if any one opens?

Answer

The whole rung loses continuity and the contactor de-energizes, even though the thermostat still calls for cooling. A technician who checks only the thermostat will miss the actual fault, since any single open interlock in that series string stops the entire circuit. This is why tracing every device on the rung matters, not just the obvious one.

#279 · Installation

How does a 24V control transformer connect into a residential control circuit, and what is terminal C for?

Answer

The transformer's primary connects to line voltage, either 120V or 240V, and its secondary delivers 24V. One secondary leg, C for common, is the return path for the entire control circuit, connecting to the C terminal at both the equipment board and the thermostat. The other secondary leg, R (or Rc/Rh), feeds the thermostat.

#280 · Installation

What happens at the thermostat and equipment when a call for cooling connects R to Y?

Answer

Connecting R to Y sends 24V to energize the compressor contactor, and on many systems the outdoor fan relay as well. This is the standard cooling call terminal in a 24V control circuit. The Y wire routes from the thermostat back to the equipment board's Y terminal, with the coil's other side returning to C.

#281 • Installation

What happens when a call for the indoor fan connects R to G?

Answer

Connecting R to G energizes the blower relay, or on an ECM-equipped system, the ECM module's fan call input. This is the standard fan call terminal in a 24V control circuit, independent of a heating or cooling call. G is commonly energized alongside Y during a cooling call or W during a heating call, depending on system design.

#282 • Installation

What is terminal W used for, and what does a heat pump's O or B terminal control?

Answer

W is the heating call terminal, connecting R to W to energize gas valves, electric heat relays, or auxiliary heat. Heat pumps add an O or B terminal for the reversing valve, energized in either heating or cooling mode depending on manufacturer convention. That convention must be confirmed before assuming which mode energizes the reversing valve.

#283 · Installation

How does a technician verify correct 24V control circuit wiring with a multimeter?

Answer

Measure roughly 24 to 30V AC across R and C at the idle thermostat first. Then confirm each call, such as Y, G, or W, produces 24V between its terminal and C at the equipment end when that call is active. This confirms both the transformer secondary and the individual call wiring are functioning correctly.

#284 · Installation

Describe a run capacitor: its rating units, typical voltage, and duty cycle.

Answer

A run capacitor is rated in microfarads (μF , often printed as MFD) for continuous AC duty, commonly at 370V or 440V. It stays connected the entire time the motor runs, shifting the phase of current in the start winding relative to the run winding to produce a rotating field from a single phase supply. Values commonly range from around $5\mu\text{F}$ for a small condenser fan up to $40\mu\text{F}$ or more for a compressor.

#285 · Installation

Describe a start capacitor: how its microfarad rating compares to a run capacitor, and its duty cycle.

Answer

A start capacitor is often five to ten times higher in microfarads than a run capacitor, but rated for intermittent duty only at a lower voltage band. It connects only during starting, through a start relay, then disconnects near running speed. Leaving it connected continuously would overheat and destroy it quickly.

#286 · Installation

Compare a run capacitor and a start capacitor in terms of duty cycle and when each is in the circuit.

Answer

A run capacitor is rated for continuous duty and stays connected the entire time the motor runs. A start capacitor is rated for intermittent duty only, connecting through a start relay just during starting and disconnecting near running speed. A start capacitor left connected continuously would overheat and fail quickly, unlike a run capacitor.

#287 · Installation

What is a dual run capacitor, and how are its two values marked?

Answer

A dual run capacitor combines two microfarad values in one can, serving both the compressor and the condenser fan from a single component. The can is marked HERM for the compressor value and FAN for the fan motor value. This is a common space-saving design in residential condensing units.

#288 · Installation

How does a technician test a run or start capacitor, and what do a zero reading versus a near-zero reading indicate?

Answer

With power disconnected and the capacitor discharged by shorting its terminals with an insulated tool, connect a multimeter's capacitance function across the terminals and compare the reading against the label value, allowing for the printed tolerance, often plus or minus 6 percent. A reading of zero means the capacitor has failed open. A reading showing continuity like a closed switch, or very low ohms, means it has shorted internally and must never be reenergized, since it can draw damaging current or rupture the can.

#289 · Installation

What does a variable frequency drive (VFD) do to an AC motor's speed?

Answer

A VFD electronically varies the frequency and voltage supplied to an AC motor, letting speed adjust smoothly instead of running only at the fixed speed set by line frequency. Internally, it rectifies fixed frequency AC input to DC, then electronically synthesizes a variable frequency AC output sent to the motor. This lets one motor run across a range of speeds instead of only on or off.

#290 · Installation

How does changing frequency above or below 60Hz affect a VFD-driven motor's speed?

Answer

Reducing frequency below the standard 60Hz slows the motor, while increasing frequency above 60Hz, within the motor's limits, speeds it up. This relationship is the basic mechanism behind a VFD's speed control. Standard 60Hz is the reference point utility power is normally supplied at in Canada.

#291 · Installation

Name two common applications of VFDs in refrigeration and HVAC equipment and their main benefit.

Answer

VFDs increasingly control compressor speed directly, matching capacity to actual load rather than cycling a fixed speed compressor on and off, improving part load efficiency substantially. They are also common on condenser fans and larger air handling fans, where reduced speed during mild weather or lower ventilation demand saves energy compared to dampers throttling a fixed speed fan. Both applications trade fixed on/off cycling for smoother, load-matched operation.

#292 · Installation

A VFD-driven compressor will not run. What should a technician check first, and what special cabling consideration applies?

Answer

Check the VFD's fault code display first, since it is usually the fastest diagnostic starting point rather than immediately suspecting the motor itself. VFD output is not the same clean sine wave as utility power, so VFD-fed motors and cables sometimes need specific cable types to manage electrical noise. A conceptual understanding of these two points is enough for most service work.

#293 · Installation

What does bonding accomplish on refrigeration and HVAC equipment, and why does it matter for safety?

Answer

Bonding connects normally non current carrying metal parts, such as an equipment frame or compressor shell, together and to the system ground. This gives a fault that energizes the frame a low resistance path back to the source, letting enough current flow to trip overcurrent protection quickly. Without bonding, a fault could leave the frame energized at a dangerous voltage instead of tripping a breaker or fuse.

#294 · Installation

How does grounding differ from bonding, even though the two are often confused?

Answer

Grounding connects the electrical system to earth, mainly for lightning and surge protection and to stabilize system voltage relative to earth potential. Bonding instead connects equipment metal parts together and to ground so a fault has a low resistance path to trip overcurrent protection. These are distinct purposes even though the two terms are frequently used interchangeably by mistake.

#295 • Installation

Under CSA C22.1, how must the bonding conductor for a piece of fixed HVAC equipment be installed?

Answer

Every piece of fixed equipment must have its exposed metal frame bonded with a conductor sized to the overcurrent device protecting the circuit, per the CEC's bonding conductor sizing tables. That conductor must be continuous, with no breaks or unauthorized splices, and it cannot rely on mechanical contact alone. CSA C22.1 is the Canadian Electrical Code that governs this requirement for equipment installed in Canada.

#296 • Installation

What does CSA C22.1 require regarding a disconnecting means for HVAC and refrigeration equipment?

Answer

A disconnecting means must be within sight of, and readily accessible to, the equipment, rated for the circuit's voltage and current. Under provincial lockout and tagout practice, it must be able to accept a physical lock so it cannot be re-energized while a technician is working. Disconnects are commonly a pull-out block, a knife blade switch, or a breaker style disconnect mounted at or near the equipment.

#297 · Installation

After opening a disconnect, why should a technician still test for zero energy before touching exposed conductors?

Answer

A mislabeled disconnect or a wiring error elsewhere can leave a supposedly de-energized circuit still live, even though the disconnect appears open. A properly rated multimeter should verify zero energy at the equipment terminals after opening the disconnect. Testing rather than trusting the disconnect's position alone is the only way to confirm the circuit is actually safe to touch.

#298 · Installation

What is the correct three-step sequence for systematic electrical troubleshooting?

Answer

Start with a visual inspection for burnt wiring, discoloured or melted terminals, a bulged capacitor, a tripped breaker, or a blown fuse. Next, perform voltage checks, starting at the disconnect and working downstream to confirm power is present where expected. Finally, with power disconnected and locked out, perform continuity or resistance checks on the suspect components this narrows down to.

#299 · Installation

Voltage is present at a contactor's line side but absent at its load side while the coil is energized. What does this indicate?

Answer

The contactor's contacts have likely failed, since the coil is pulling in but current is not passing through to the load side. This differs from voltage being absent at the line side itself, which would instead point to a fault upstream, such as a blown fuse. Voltage checks worked in sequence from source to load isolate exactly where in the circuit the fault sits.

#300 · Installation

Why might a resistance check catch a compressor winding fault that a simple continuity check would miss?

Answer

A continuity check only shows whether a path is open, reading infinite resistance or OL on many digital meters, or complete, not whether the resistance value is normal. A resistance check compares the measured value against specification or a known good component, which can reveal a partially shorted or open winding that still shows some continuity. This makes resistance checks especially useful on compressor, transformer, and motor windings.

#301 · Commissioning

Why should every terminal in the disconnect, contactor, and compressor terminal box be physically checked with a screwdriver at startup, not just visually inspected?

Answer

Vibration during shipping and handling loosens factory electrical connections even when they look intact. A loose terminal can arc and destroy a compressor winding within the first thirty seconds of run time, so mechanical tightness must be confirmed by hand, not by sight alone.

#302 · Commissioning

Why must three-phase equipment with a scroll compressor be checked with a phase rotation meter before startup?

Answer

Scroll compressors are direction sensitive. Wired backward, a scroll compressor runs hot and loud but never develops a real pressure differential, so confirming correct phasing before energizing prevents a startup that looks like it is running but is not actually pumping refrigerant.

#303 • Commissioning

What is the typical acceptable tolerance between supply voltage and compressor nameplate voltage before startup?

Answer

Supply voltage should match nameplate voltage within about plus or minus 10 percent. A reading outside that band should be investigated and corrected before the compressor is brought online, since sustained under voltage or over voltage stresses the motor windings.

#304 • Commissioning

What position should king valves and compressor service valves be in for normal operation before startup, and why does gauge port capping matter?

Answer

They should be fully back-seated for normal operation with gauge ports capped. A valve left cracked or a port left uncapped can leak refrigerant continuously or send liquid the wrong direction, so the pre-startup walkthrough confirms every valve position, not just that the valves exist.

#305 • Commissioning

What eventually happens to a refrigerant line set that is not supported at code-required intervals?

Answer

An unsupported line set vibrates continuously during operation. Over time that vibration work-hardens the metal at the joints, and the line eventually fractures, so proper support spacing is confirmed visually during the pre-startup walkthrough.

#306 • Commissioning

During pre-startup verification, what specifically must be confirmed about high and low pressure cutouts, beyond their physical presence in the panel?

Answer

The cutouts must actually be wired into the control circuit, not merely installed in the panel. A safety device sitting unwired will not stop the compressor when a dangerous pressure condition occurs, so wiring continuity is checked, and the high pressure setpoint is confirmed against the refrigerant in use.

#307 · Commissioning

What is the typical minimum crankcase heater warm-up time before first startup for larger commercial compressors versus smaller residential units?

Answer

Larger commercial compressors typically need a minimum of 12 to 24 hours of crankcase heater warm-up, while smaller residential units need several hours, always per manufacturer instructions. This warm-up must occur before line voltage is applied for the first startup or after any extended period without power.

#308 · Commissioning

Why is a micron gauge used to confirm evacuation instead of the compound gauge on the manifold set?

Answer

The compound manifold gauge loses meaningful resolution long before the vacuum gets deep enough to boil off residual moisture. A micron gauge is the only field instrument accurate enough to confirm the system reached the deep vacuum target required before charging.

#309 · Commissioning

What vacuum level do CSA and manufacturer specifications typically require before charging a standard commercial or residential system?

Answer

The system should be pulled down to about 500 microns, roughly 0.5 mmHg or about 0.02 inHg vacuum. Larger commercial or low-temperature systems often carry a tighter target than this standard 500-micron benchmark.

#310 · Commissioning

Once the target vacuum is reached, why should the vacuum pump be isolated with a valve rather than simply switched off?

Answer

A pump left connected while shutting down lets air migrate backward through its oil as the pump cools. Isolating the pump at the core tool or a manifold valve seals the system off cleanly so the rate-of-rise test reflects the system alone, not the pump.

#311 · Commissioning

During a rate-of-rise vacuum test, the micron reading climbs from 500 to about 1,200 microns, then levels off and holds. What does this indicate?

Answer

A rise that levels off and holds after climbing is normal residual moisture outgassing, not a leak. The system can proceed to charging once the reading stabilizes, since this pattern is expected on nearly every properly evacuated system.

#312 · Commissioning

During a rate-of-rise vacuum test, the micron reading climbs steadily without leveling off, continuing back toward atmospheric pressure. What does this indicate?

Answer

A reading that keeps climbing without leveling off indicates an actual leak in the system, not moisture outgassing. The leak must be located and repaired, and the system re-evacuated, before any refrigerant is introduced.

#313 • Commissioning

Name at least five measurements that belong in a complete baseline commissioning data record besides suction and discharge pressure.

Answer

A complete baseline record also includes suction line temperature and calculated superheat, liquid line temperature and calculated subcooling, compressor amp draw on each leg against nameplate rated load amps, voltage at the compressor terminals under load, ambient outdoor and return air temperature, and supply air temperature split across the coil. The date, mechanic's name, and conditions present at the time are recorded alongside the readings themselves.

#314 • Commissioning

When is a triple evacuation with nitrogen breaks between pulls recommended instead of a single standard vacuum pull?

Answer

It is recommended whenever moisture contamination is suspected, such as after a compressor burnout recovery or on a system left exposed to atmosphere for an extended period. Breaking nitrogen between pulls, or pulling a single extended vacuum well below 500 microns, gives far higher confidence than one quick pump-down on a contaminated system.

#315 • Commissioning

When is weighing in a refrigerant charge by mass the correct charging method, rather than superheat or subcooling?

Answer

Weigh-in is used whenever the equipment nameplate or installation manual specifies an exact charge weight, typically on a factory-charged split system near its base line set length, or after a full recovery where a known quantity must be replaced. It is the most accurate method available when a specified target mass exists.

#316 • Commissioning

How is a factory-specified charge weight adjusted for a split system whose line set is longer than the manufacturer's base length?

Answer

The mechanic adds a manufacturer-specified correction, commonly expressed as a certain number of grams per metre of additional liquid line beyond the base length. The scale under the charging cylinder is zeroed first so the mechanic can charge to the total corrected mass accurately.

#317 • Commissioning

The superheat method for verifying charge applies to which type of metering device, and why?

Answer

It applies to fixed-orifice metering devices, meaning piston or capillary tube systems with no active feedback control over refrigerant flow. Because the orifice cannot respond to load changes, correct charge is defined by hitting a target superheat cross-referenced against a manufacturer chart for the actual conditions.

#318 • Commissioning

State the formula for calculating superheat at the evaporator outlet, and identify the two measurements it requires.

Answer

Superheat equals the actual suction line temperature minus the saturation temperature. The line temperature comes from a clamp-on probe at the evaporator outlet, and the saturation temperature comes from converting measured suction pressure using a pressure-temperature chart for the refrigerant in use.

#319 • Commissioning

On a fixed-orifice system, measured suction pressure converts to a saturation temperature of 2°C. The clamp-on probe at the evaporator outlet reads 9°C. Calculate superheat.

Answer

Superheat equals actual line temperature minus saturation temperature, so 9°C minus 2°C equals 7°C of superheat. That figure is then compared against the manufacturer's superheat chart for the current indoor and outdoor conditions to judge whether the charge is correct.

#320 • Commissioning

If a measured superheat reading is lower than the manufacturer's target, is the system overcharged or undercharged, and what corrects it?

Answer

A lower than target superheat means the system is overcharged, since adding refrigerant lowers superheat by letting more liquid reach the coil before it fully boils off. The correction is to remove refrigerant in small amounts until superheat rises back to the target.

#321 · Commissioning

The subcooling method for verifying charge applies to which type of metering device, and why is superheat not used there instead?

Answer

It applies to systems using a thermostatic expansion valve (TXV) or electronic expansion valve (EXV). Because these devices actively regulate superheat at the evaporator outlet regardless of charge level within a reasonable range, superheat readings become useless for judging charge, so subcooling at the condenser outlet is used instead.

#322 · Commissioning

State the formula for calculating subcooling at the condenser outlet, and identify the two measurements it requires.

Answer

Subcooling equals the saturation temperature minus the actual liquid line temperature. The saturation temperature comes from converting measured high side liquid line pressure using a pressure-temperature chart, and the actual temperature comes from a clamp-on probe at the condenser outlet.

#323 · Commissioning

An R-410A TXV system shows liquid line pressure converting to 45°C saturation temperature, with the liquid line thermometer reading 34°C. Calculate subcooling.

Answer

Subcooling equals saturation temperature minus actual line temperature, so 45°C minus 34°C equals 11°C of subcooling. Against a manufacturer target of 8°C, this reading is 3°C high, indicating the system is overcharged.

#324 · Commissioning

On a TXV system with an 8°C subcooling target, a stable reading comes back at 11°C. What corrective action does the mechanic take, and how?

Answer

Because the system is overcharged, the mechanic recovers small amounts of refrigerant in stages, letting the system stabilize between each removal. Refrigerant is removed gradually until subcooling reads 8°C at stable operating conditions, avoiding overcorrection from removing too much at once.

#325 · Commissioning

On a TXV system with an 8°C subcooling target, a stable reading comes back at 4°C. What does this indicate, and how is it corrected?

Answer

A 4°C reading is 4°C below the 8°C target, indicating the system is undercharged. The correction runs opposite to the overcharge case: the mechanic adds refrigerant in small metered amounts, letting the system stabilize between additions, until subcooling climbs to 8°C.

#326 · Commissioning

List the order of steps a mechanic follows once crankcase warm-up, oil, and pre-startup checks are complete, up to compressor startup.

Answer

The mechanic confirms disconnect is on and control voltage is present, confirms the thermostat or controller is calling correctly, then brings the compressor online while watching suction and discharge pressures for a normal rise and stabilization pattern. The mechanic also listens for abnormal noise in the first thirty seconds and confirms the condenser fan and evaporator blower are both running in correct rotation and airflow direction.

#327 · Commissioning

When is charging refrigerant as a vapour into the low side suction line the correct and safe practice?

Answer

Vapour charging into the suction line is standard for small trim adjustments on a system that is already running, because the vapour is simply compressed along with the rest of the suction gas. It carries no risk of liquid slugging as long as the compressor is actively pulling refrigerant through the line.

#328 · Commissioning

When is charging refrigerant as a liquid required rather than vapour, and through which service port?

Answer

Liquid charging is required for a large initial charge into an empty or nearly empty system, and it is done through the high side liquid line or king valve port with the compressor off. It is also faster than vapour charging for large quantities.

#329 · Commissioning

Why must liquid refrigerant never be charged into the suction side of a compressor that is not running?

Answer

Liquid entering the suction line of a stopped compressor pools in the crankcase instead of flashing to vapour. The instant that compressor starts, the pooled liquid causes a slugging event, which can bend connecting rods or shatter valve plates on the very first startup.

#330 · Commissioning

What is a zeotropic blend, and name three examples used in refrigeration and air conditioning.

Answer

A zeotropic blend is a mixture of refrigerant components with different boiling points, which causes it to exhibit temperature glide rather than boiling or condensing at a single fixed temperature. R-410A, R-407C, and R-448A are all zeotropic blends.

#331 • Commissioning

What is fractionation, and why does it happen when a zeotropic blend is charged as a vapour off the top of the cylinder?

Answer

Fractionation is the preferential evaporation of the lower-boiling-point component of a blend, leaving the cylinder's remaining refrigerant with a different composition than what left the factory. Charging vapour draws off the lighter component first, so the last portion of that cylinder throws off the system's designed pressure-temperature relationship and capacity.

#332 • Commissioning

Regardless of whether a system's charging point is high side or low side, how must every cylinder of zeotropic blend refrigerant be charged?

Answer

Every cylinder of zeotropic blend refrigerant must be charged as a liquid, drawn from the bottom of an upright cylinder or through a dip tube. A dedicated liquid charging device, or simply inverting a cylinder without a dip tube, are both accepted field methods to guarantee a liquid draw and avoid fractionation.

#333 · Commissioning

Describe the correct technique for purging air from a manifold gauge hose before connecting it to a system's refrigerant circuit.

Answer

With the gauge set connected to the refrigerant source and the service port, the mechanic briefly cracks the appropriate manifold valve to let a short burst of refrigerant push air out through the far hose's Schrader depressor fitting, directed away from the technician. The valve is then closed immediately, since skipping this step introduces air and moisture into the system with every connection made.

#334 · Commissioning

What advantage does a digital gauge set have over an analog gauge set during charging?

Answer

A digital gauge set lets the refrigerant type be selected electronically and displays saturation temperature directly alongside pressure, removing the risk of reading the wrong refrigerant scale on an analog dial. When temperature probes are connected, it also calculates superheat and subcooling automatically, reducing the chance of a manual chart-reading error.

#335 • Commissioning

What should be done with manifold gauge hoses at the end of a charging procedure, and under what regulatory framework?

Answer

Hoses are recovered into the low side briefly before disconnecting, minimizing the refrigerant trapped in the hoses that would otherwise vent to atmosphere. This practice is consistent with regulations under the Canadian Environmental Protection Act and provincial equivalents governing refrigerant handling.

#336 • Commissioning

During charging, which manifold valve is opened when trimming vapour into a running system, and which is opened when charging liquid into an off system?

Answer

The low side valve is opened when trimming vapour into a running system's suction line, and the high side valve is opened when charging liquid into an off system's liquid line. Both manifold valves stay closed except for whichever one is actively being used for the procedure underway.

#337 • Commissioning

On a fixed-orifice piston system, what happens if the installed piston is undersized versus oversized for the equipment tonnage?

Answer

An undersized piston starves the evaporator, producing high superheat and low system capacity. An oversized piston floods the evaporator, producing low superheat with a real risk of liquid returning to the compressor, so the correct piston size must be confirmed against the manufacturer's sizing chart at startup.

#338 • Commissioning

If a piston system's charge is correct by weight but superheat still reads out of spec, what should the mechanic suspect first, and why?

Answer

Piston systems have no field adjustment beyond the fixed orifice size itself. An out-of-spec superheat reading on a properly charged system usually points back to airflow, a wrong piston size, or a restriction elsewhere in the circuit, not the charge.

#339 · Commissioning

On a TXV with a field-adjustable superheat setting, which direction does the mechanic turn the stem to raise target superheat, and which to lower it?

Answer

Turning the stem clockwise in small increments, typically a quarter turn, raises the target superheat. Turning it counterclockwise lowers the target superheat, and the seal cap is removed and replaced using the small tool supplied with the valve.

#340 · Commissioning

Why does a mechanic wait ten to fifteen minutes between each TXV adjustment instead of making several quick turns in a row?

Answer

TXVs respond slowly to a stem adjustment, so the system needs time to restabilize before the effect of one change can be accurately judged. Overcorrecting from impatience, by adjusting again before the previous change has settled, is a common startup mistake.

#341 • Commissioning

On a properly subcooled TXV system, what is the typical target superheat range once the valve is correctly adjusted?

Answer

Target superheat on a properly subcooled TXV system typically falls between 5°C and 8°C, roughly 9°F to 14°F. Once subcooling and superheat both land in range together, the system is considered correctly charged with the valve correctly adjusted.

#342 • Commissioning

Compare polyolester (POE) oil and mineral oil use in refrigeration compressors, and explain why the two must never be mixed.

Answer

POE oil is used with HFC refrigerants such as R-410A and R-407C, while mineral oil remains compatible with older HCFC systems. Mixing the two is never acceptable because POE oil is highly hygroscopic and reacts differently with system materials than mineral oil does, so the correct oil type must match the compressor manufacturer's specification exactly.

#343 • Commissioning

What does a crankcase heater actually prevent, and what happens to a compressor started without adequate warm-up time?

Answer

The crankcase heater keeps compressor sump oil warmer than the surrounding refrigerant, preventing refrigerant vapour from migrating into the oil and condensing there during the off cycle. Without warm-up, dissolved refrigerant in a cold sump flashes to vapour violently at startup, foaming the oil and causing liquid slugging in the cylinders or scroll set, which destroys more new compressors during commissioning than almost any other mistake.

#344 • Commissioning

What is the approximate conversion between CFM and L/s, and which unit is still most commonly called out on a Canadian jobsite?

Answer

1 CFM equals roughly 0.47 L/s. Even though L/s is the proper SI unit and appears on newer drawings, CFM remains the number techs call out to each other in the field, for example over a two-way radio during a rooftop unit test.

#345 • Commissioning

Compare a rotating vane anemometer to a hot wire anemometer for grille airflow readings.

Answer

A rotating vane anemometer measures velocity directly at a grille and that reading is multiplied by the grille's free area to get CFM. A hot wire anemometer performs the same job but responds better at low velocities, which matters at wide-throw diffusers where a vane anemometer can stall.

#346 • Commissioning

A tech reads grille velocity with a vane anemometer but has no free area value for the grille. Why is the reading useless?

Answer

CFM is calculated by multiplying velocity by the grille's effective free area, so without that area figure the velocity number cannot be converted into an airflow value. The free area is usually printed on the manufacturer's data plate or found by looking up the model number.

#347 · Commissioning

What is a duct traverse and when is it preferred over a single grille reading?

Answer

A duct traverse uses a pitot tube connected to a manometer to take velocity pressure readings across a grid of points on a straight duct run, then averages them and applies the cross-sectional area for a total CFM figure. It is far more reliable than a single-point reading taken near a fitting, elbow, or damper where airflow is turbulent.

#348 · Commissioning

Name the two standard grid methods used to lay out pitot tube traverse points across a duct cross-section.

Answer

The log-Tchebycheff method and the equal-area method are the two standard traverse grid layouts. Both give the tech a defined set of points to sample so the averaged velocity pressure represents the whole cross-section rather than one arbitrary spot.

#349 · Commissioning

What does total external static pressure (TESP) tell a mechanic, and what unit does a Canadian commissioning report typically use for it?

Answer

TESP shows how hard the fan is working to push air through the ductwork, filters, coils, and diffusers, and it is measured with a manometer connected to a static pressure tip. Canadian commissioning reports increasingly specify pascals (Pa) as the primary unit rather than inches of water column.

#350 · Commissioning

TESP on a rooftop unit reads higher than the design value on the fan curve. What are the most common causes?

Answer

A high TESP reading usually means something downstream is restricting airflow. Common causes include a dirty filter, a closed or partially closed damper, or an undersized duct run creating extra resistance.

#351 · Commissioning

TESP reads lower than design on a commissioned unit. What should the mechanic suspect?

Answer

A lower than design TESP suggests the fan is not seeing the resistance it was sized to overcome. This points to a duct leak or a damper stuck open that is bypassing part of the system's intended resistance.

#352 · Commissioning

Explain the proportional balancing method for adjusting supply air dampers across a zone.

Answer

The tech balances the branch closest to the fan first, then works outward toward the farthest branches. This produces a more stable result than chasing individual outlets randomly, because closing any one damper shifts pressure and airflow to the rest of the system.

#353 • Commissioning

Why does a mechanic need to confirm which flow unit a hydronic drawing set uses before recording balancing numbers?

Answer

Hydronic flow is measured in either GPM (US gallons per minute) or L/s depending on the project, and the two units are not interchangeable without conversion. Recording readings in the wrong assumed unit would make the whole balancing report inaccurate.

#354 • Commissioning

What is a calibrated balancing valve (circuit setter) and how does it produce a flow reading?

Answer

A calibrated balancing valve has a built-in venturi or orifice with pressure ports on either side of the restriction. A differential pressure reading across those ports is converted to a flow rate using the valve manufacturer's chart or an electronic balancing meter.

#355 · Commissioning

Compare balancing a two-pipe reverse-return hydronic system to balancing a direct-return system.

Answer

A reverse-return system is more forgiving because its piping geometry is already designed to equalize resistance across branches. A direct-return system usually needs more aggressive valve adjustment at the near end to create enough resistance that the far end still receives its full design flow.

#356 · Commissioning

A chilled water coil is designed for a 5.5°C delta-T but shows only 2°C under full load. What does this usually indicate?

Answer

A delta-T well below design under full load usually means flow through that coil is too high relative to its capacity. That excess flow robs delta-T from the rest of the loop and forces the chiller to work harder than necessary.

#357 · Commissioning

Why does the proportional balancing approach require re-checking earlier hydronic settings as later terminals are adjusted?

Answer

Adjusting a valve at one terminal changes the pressure differential across the whole loop, which shifts flow at terminals that were already set. Re-checking earlier settings after later ones are adjusted is necessary to keep the whole loop at its design flow values.

#358 · Commissioning

Before performance verification testing begins, what documents should the mechanic have in hand?

Answer

The mechanic needs the original mechanical drawings, the equipment schedule, and the manufacturer's performance data. These give the design values that measured readings will be compared against, rather than a generic rule of thumb.

#359 · Commissioning

List the key measurements compared against the equipment schedule during a refrigeration system performance verification.

Answer

Comparisons include suction and discharge pressures converted to saturation temperatures, superheat and subcooling, entering and leaving air or water temperatures at the evaporator and condenser, and total capacity calculated from measured airflow and coil temperature split. Every figure is checked against the equipment schedule's stated design condition.

#360 · Commissioning

Why is it a mistake to compare a system's readings on a mild day directly against its full design load numbers?

Answer

A system running at partial load on a cool day will naturally show different readings than one operating at full design load. Comparing readings taken under different operating conditions against the full-load design numbers leads to false conclusions about system performance.

#361 • Commissioning

A compressor motor is pulling amperage close to its nameplate FLA rating even though the system is at partial load. What does this signal?

Answer

Amperage that is high relative to the actual load usually points to a problem such as a mechanical binding issue, incorrect belt tension, or a refrigeration side fault forcing the compressor to work harder than the load requires. It should be checked against nameplate FLA and the design engineer's expected running amps.

#362 • Commissioning

Measured discharge air temperature does not match the design value. List the possible root causes a mechanic should work through.

Answer

Possible causes include an incorrect refrigerant charge, airflow across the coil being too low, the outdoor test condition not matching the equipment schedule's rating condition, or the equipment having been sized incorrectly at the design stage. Documenting which cause applies, with supporting readings, is what separates real verification from just recording numbers.

#363 • Commissioning

Why can a pressure gauge that reads 35 kPa low be worse than an obviously broken gauge?

Answer

A gauge that is off by a fixed amount can either send a mechanic chasing a problem that does not exist or hide a real problem because the readings happen to look acceptable on paper. An obviously broken gauge, by contrast, gets replaced immediately rather than trusted.

#364 • Commissioning

What calibration interval do most instrument manufacturers specify as a minimum for field instruments like manifold gauges and manometers?

Answer

Most manufacturers specify an annual calibration interval as a minimum. Some critical instruments need more frequent checks depending on how hard they are used, since mechanical shock, temperature extremes, and sensing element age all cause drift.

#365 · Commissioning

Describe a quick field check a mechanic can run to verify a pressure gauge before connecting it to a system.

Answer

The gauge should read zero at atmospheric pressure before it is connected to any system. This simple check does not replace a proper calibration certificate but catches a badly drifted gauge before it produces readings that end up in a formal report.

#366 · Commissioning

What two reference points can a mechanic use in the field to check a temperature probe without lab equipment?

Answer

An ice bath provides a known 0°C reference point, and boiling water provides a known 100°C reference point at sea level, adjusted for altitude. Checking the probe against both catches a significantly drifted sensor before it is trusted for reporting.

#367 · Commissioning

What instrument information must a commissioning report record, and why does this matter for a later warranty dispute?

Answer

The report should list the make, model, and serial number of every instrument used, along with the date of its most recent calibration. This protects the technician by demonstrating the readings were collected with instruments known to be accurate at the time of the test, not whatever gauge happened to be in the truck.

#368 · Commissioning

Why is sound measured on the A-weighted scale (dBA) rather than an unweighted decibel scale during commissioning?

Answer

The A-weighted scale approximates how the human ear actually perceives loudness across different frequencies. Readings are taken at specified locations, often the centre of an occupied space at standard height, and compared against the project's stated NC or RC rating.

#369 · Commissioning

An office space has a design sound target of NC 30 to NC 35. Does this target also apply to the mechanical room?

Answer

No. A mechanical room has no such restriction because it is not an occupied space, unlike the office area where the NC or RC target is meant to protect occupant comfort.

#370 · Commissioning

List the common root causes of a vibration complaint on refrigeration or air conditioning equipment.

Answer

Common causes are an unbalanced fan or compressor, a coupling misalignment, a failed or improperly selected vibration isolator, or resonance where the equipment's operating frequency matches a natural frequency of the supporting structure. A vibration meter measures velocity or acceleration at key points to identify which is at fault.

#371 · Commissioning

A rooftop unit is well balanced mechanically but the building still reports vibration transmitted into the structure below. What should the mechanic inspect first?

Answer

The mechanic should inspect the isolation devices, since a spring isolator that is bottomed out or a rubber mount that has hardened with age will transmit vibration directly into the structure. No amount of adjusting the equipment itself will fix a problem that is actually in the isolation.

#372 · Commissioning

What is the sequence of operations, and why can hardware that works perfectly still fail commissioning?

Answer

The sequence of operations is the written logic describing how the system should behave under every condition, including startup, normal operation, unoccupied setback, alarm conditions, and safety shutdowns. If the controls do not execute that logic correctly, the system fails commissioning even if every piece of hardware is functioning.

#373 • Commissioning

Give three examples of safety interlocks a mechanic must verify while testing sequence of operations.

Answer

Examples include a low pressure cutout actually stopping the compressor, a high discharge temperature limit actually shutting down the unit, and a proof-of-flow switch actually preventing compressor startup when water flow is not established. Each interlock has to be forced into its trigger condition to confirm it responds correctly.

#374 • Commissioning

A BAS graphic shows a different supply air temperature than the equipment's own controller display. What does this mismatch usually mean?

Answer

A mismatch between BAS points and the equipment's own readings usually means a sensor calibration issue or a wiring or addressing error at the BAS integration point. It needs to be resolved before handover, since a facility operator relying on inaccurate BAS data will make wrong decisions about the equipment.

#375 · Commissioning

Why are BAS trend logs valuable evidence in a commissioning report beyond a single spot-check reading?

Answer

A trend log, such as supply air temperature holding steady through a full occupied cycle or a chiller cycling at a reasonable frequency, gives documented proof the system behaves correctly over time. A single gauge reading only proves performance at the moment the mechanic happened to be standing there.

#376 · Commissioning

List the minimum contents required in a proper commissioning report.

Answer

At minimum a report needs the project and equipment identification, the date and outdoor conditions tests were performed under, every instrument used with its calibration date, the design values from the drawings and equipment schedule, the actual measured values, and a clear pass or fail determination against a stated tolerance for each test point. Photographs of nameplate data, gauge readings, and observed defects strengthen the report further.

#377 · Commissioning

Why does outdoor temperature at the time of testing matter enough to record in a commissioning report?

Answer

Capacity comparisons depend on the operating condition the system was tested under, and outdoor temperature directly affects that condition. Recording it lets anyone reviewing the report later judge whether the measured numbers should match full design load or a partial load condition.

#378 · Commissioning

What four pieces of information should every entry on a deficiency list include?

Answer

Each entry should describe the specific location, the nature of the deficiency, the trade responsible for correcting it, and a target date for resolution. A deficiency list is separate from the commissioning report but itemizes anything that does not meet design intent or code requirement.

#379 · Commissioning

Compare a vague deficiency entry to a properly written one for a disconnected flex duct connection at a rooftop unit.

Answer

A vague entry like "duct leaking somewhere near unit 3" gives almost nothing to act on. A proper entry states the specific location, the nature of the defect, and the impact, for example: flexible duct connection at RTU-3 supply, disconnected clamp, approximately 15 percent of measured design airflow lost, correction required before final balance can be verified.

#380 · Commissioning

A client questions months after handover whether a system was ever performing correctly. How does a proper commissioning report protect the mechanic?

Answer

A commissioning report with dated, instrument-verified readings is the mechanic's evidence that the system was performing to specification at the time of handover. Without that paper trail, a warranty or liability dispute becomes a matter of one party's word against the other's.

#381 • Commissioning

List the minimum topics client training should cover at handover.

Answer

Training should cover operating the thermostat or BAS interface for day-to-day adjustments, the routine maintenance schedule such as filter changes and belt inspection, what a normal versus an alarm condition looks like, and who to call for service versus what the operator can safely handle. The level of detail should be tailored to the audience.

#382 • Commissioning

Why should a mechanic walk a client through the O&M; manual rather than just leaving it on a desk?

Answer

Assuming the manual will be read cover to cover is unrealistic, so the mechanic should point out where key information actually lives. This includes the location of shutoff valves and disconnects, warranty terms and what voids them, and the manufacturer's recommended maintenance intervals.

#383 · Commissioning

What four items make up a complete handover package at project close-out?

Answer

A complete handover package includes the commissioning report, the deficiency list with sign-off on completed corrections, the O&M; manuals, and a training record showing who was trained and on what date. Together these close out the project professionally.

#384 · Commissioning

A facility operator has years of mechanical experience while an office manager mainly uses a thermostat. Should their handover training be identical?

Answer

No. Training should be tailored to the audience, so the mechanically experienced operator needs a deeper level of detail on system operation and maintenance than the office manager, who mainly needs to know normal thermostat use and who to call for service.

#385 • Commissioning

Why does treating client handover with the same care as technical fieldwork matter for a Red Seal mechanic's career?

Answer

A thorough, professional handover closes out the project cleanly and demonstrates competence beyond just the technical work. This builds the kind of reputation that generates repeat business and referrals, which matters as much in this trade as the ability to properly charge a system or balance a duct run.

#386 • Maintenance and Service

Why does a written preventive maintenance schedule matter more than a one-time inspection?

Answer

A schedule assigns a task, a frequency, and an acceptable range of readings so every visit can be compared against a baseline. This lets a technician see drift toward failure before it actually fails. Without a baseline, every visit is a fresh guess instead of a trend.

#387 • Maintenance and Service

What are the typical filter inspection and replacement intervals for a commercial rooftop unit versus a residential system?

Answer

A commercial rooftop unit typically needs monthly filter inspection and replacement every one to three months. A residential fiberglass filter is usually checked every one to two months. Dusty or greasy environments like restaurants and sawmills shorten these intervals considerably.

#388 • Maintenance and Service

How does a dirty condenser coil affect head pressure and compressor amperage?

Answer

A condenser coil caked with dust, cottonwood fluff, or grass clippings cannot reject heat efficiently. This drives head pressure and compressor amperage up together. Coils should be inspected each visit and cleaned at least once a year, twice in high-debris locations.

#389 • Maintenance and Service

How does a dirty or biologically fouled evaporator coil affect system performance?

Answer

A coil coated in dust or biological growth loses heat transfer surface area. This starves the refrigerant of the heat it needs to boil off inside the coil. The reduced heat transfer lowers cooling capacity even though airflow may still seem adequate at a glance.

#390 • Maintenance and Service

Why can a slack drive belt cause airflow problems that are not visually obvious?

Answer

A belt that has stretched and gone slack slips under load, cutting airflow. There is no obvious visual sign until the belt is actually touched and deflected by hand. This applies to any unit using a belt-driven blower or belt-driven condenser fan.

#391 • Maintenance and Service

Why are bearing checks for play, noise, and lubrication part of a preventive maintenance visit?

Answer

A seized bearing can lock a motor shaft and burn out the windings within minutes. Checking bearings on a set schedule catches wear before it causes a sudden, expensive failure. This applies to both blower motors and condenser fan motors.

#392 • Maintenance and Service

Trace the failure chain that starts from a single skipped filter change.

Answer

A dirty filter causes low airflow, low airflow causes a cold or frosted evaporator coil, and a frosted coil restricts airflow further in a feedback loop. Eventually liquid refrigerant reaches the compressor. What starts as a cheap filter becomes a compressor replacement if the chain is not broken early.

#393 · Maintenance and Service

Discharge pressure and saturation temperature sit above the expected condensing temperature for the ambient. What fault is this, and what causes it?

Answer

This is a high head pressure fault, which almost always points to a condenser that cannot reject heat properly. Common causes are a dirty or blocked condenser coil, a failed or slow condenser fan motor, condenser air recirculation, an overcharged system, or non-condensable gases from a poor evacuation. Suction side readings usually stay closer to normal unless the fault has progressed far enough to affect the whole circuit.

#394 · Maintenance and Service

What does a preventive maintenance contract typically include, and how does it benefit both the client and the contractor?

Answer

A contract covers a fixed number of visits per year with a defined task list for each visit. The client gets predictable costs and fewer emergency failures. The contractor gets steady work and the chance to catch small issues during a planned visit instead of an emergency call.

#395 • Maintenance and Service

List the common causes of low suction pressure on a vapour compression system.

Answer

Common causes are a low refrigerant charge, a restricted metering device, a plugged filter drier, a frozen evaporator coil, or low airflow across the evaporator. All of these reduce the mass of refrigerant reaching the compressor, so suction pressure drops. The technician narrows the cause by checking subcooling and looking for a temperature drop across the drier and metering device.

#396 • Maintenance and Service

How does a technician distinguish a low refrigerant charge from a restriction when suction pressure is low?

Answer

Low subcooling combined with low suction pressure suggests an undercharge. A restriction is confirmed instead by inspecting the drier and metering device for a temperature drop across them, since a restriction causes that drop even when the charge itself is correct. Both checks are needed because low suction pressure alone does not separate the two.

#397 • Maintenance and Service

What defines short cycling, and what is a normal compressor run cycle for comparison?

Answer

Short cycling is a compressor turning on and off far more often than its designed run cycle, every few minutes instead of a normal 10 to 20 minute cycle. It is hard on the compressor because the highest electrical stress happens at startup, and frequent starts multiply that stress over the unit's life. It always has an underlying cause that must be traced, not just accepted.

#398 • Maintenance and Service

A compressor is short cycling every few minutes. What are the likely causes a technician should check?

Answer

Likely causes include an oversized unit for the load, a poorly placed thermostat that satisfies quickly, a dirty filter tripping a safety, a low pressure switch cutting out on marginal charge, or a failing run capacitor causing a hard start that trips the overload. Each cause is checked in turn since the symptom looks identical regardless of which one is present.

#399 • Maintenance and Service

At what coil surface temperature does frost or ice begin to form on an evaporator, and what usually causes it?

Answer

Frost or ice forms when coil surface temperature drops below 0 degrees C for an extended period. The most common causes are low airflow, low charge, or a metering device feeding too little refrigerant to the coil. A dirty filter or blocked return is the single most common airflow cause.

#400 • Maintenance and Service

What is the correct diagnostic order for tracing an evaporator frost or ice buildup complaint?

Answer

A technician checks airflow first, since a dirty filter or blocked return is the most common single cause. Suction pressure and superheat are checked next to separate an airflow-only problem from a charge or metering fault. Jumping straight to charge adjustment without checking airflow first risks masking the real problem.

#401 • Maintenance and Service

For a frosted coil, what does low suction pressure with high superheat indicate, versus low suction pressure with normal superheat and good airflow?

Answer

Low suction pressure with high superheat suggests an undercharge or a restriction starving the coil of refrigerant. Low suction pressure with low or normal superheat, combined with confirmed good airflow, points toward an airflow problem alone rather than a charge or metering issue. Superheat is the key value that separates the two otherwise similar-looking symptoms.

#402 • Maintenance and Service

Describe the pressure, temperature, and superheat signature of refrigerant flood-back to the compressor.

Answer

Flood-back shows low superheat, sometimes at or near zero, with a suction line cold and sweating well past the compressor inlet. In severe cases oil foaming is visible in a sight glass. This is the opposite pattern from a starved evaporator, which shows high superheat instead.

#403 · Maintenance and Service

What conditions cause liquid refrigerant to flood back to the compressor?

Answer

Flood-back is caused by an overcharged system, a metering device stuck open, a failed or missing crankcase heater on a system prone to migration, or very low load conditions where the evaporator cannot boil off refrigerant fast enough. Any of these lets liquid reach the suction line instead of fully vaporized vapour.

#404 · Maintenance and Service

What compressor damage results if refrigerant flood-back is left uncorrected?

Answer

Flood-back washes oil out of the compressor over time, robbing bearings of lubrication. It also causes liquid slugging, which damages bearings and valves directly on each stroke. Left long enough it condemns the compressor entirely.

#405 · Maintenance and Service

Describe the pressure, temperature, and superheat signature of a starved evaporator.

Answer

A starved evaporator produces high superheat with a suction line that stays warm closer to the coil inlet. Cooling capacity is reduced even though the compressor runs, pulling low suction pressure as a result. This is the mirror image of flooding, which shows low superheat and higher suction pressure instead of high superheat and low suction pressure.

#406 · Maintenance and Service

What causes an evaporator coil to be starved of refrigerant?

Answer

Causes include a plugged or undersized metering device, a plugged filter drier, a low charge, or a liquid line restriction such as a partially closed service valve. Any of these reduces the flow of refrigerant reaching the coil below what the load requires.

#407 • Maintenance and Service

How does a technician confirm that an evaporator is starved rather than simply undercharged?

Answer

The technician measures temperature drop across the filter drier and the metering device. A healthy drier shows almost no measurable temperature drop across it, while a restricted one shows a marked drop. This pinpoints a physical restriction even when the overall system charge could otherwise appear correct.

#408 • Maintenance and Service

A technician measures low suction pressure and unusually high superheat at the evaporator outlet. What fault does this point to?

Answer

This signature points to a starved evaporator, meaning too little refrigerant is reaching the coil. Likely causes are a plugged or undersized metering device, a plugged filter drier, a low charge, or a restricted liquid line. Confirming which one requires checking for a temperature drop across the drier and metering device.

#409 • Maintenance and Service

A technician measures unusually high suction pressure with superheat near zero. What fault does this point to?

Answer

This signature points to refrigerant flooding back to the compressor rather than a starved coil. Likely causes are an overcharge, a metering device stuck open, a failed or missing crankcase heater, or very low load conditions. Left uncorrected this risks liquid slugging and bearing or valve damage.

#410 • Maintenance and Service

What information should be logged on every preventive maintenance service record?

Answer

Each visit should log the date, readings taken including suction and discharge pressure, superheat, subcooling, and amperage draw, any refrigerant added or recovered with quantity and type, and any parts replaced. This lets the next technician see a trend line instead of a single snapshot. Complete records also document that leak repairs met regulatory reporting obligations.

#411 · Maintenance and Service

Describe how a winding resistance test is performed on a de-energized compressor.

Answer

An ohmmeter is used across the common, run, and start terminals to check for a short, an open, or a ground fault. The unit must be de-energized and, ideally, refrigerant pressure equalized before testing. Readings between terminals are compared against the compressor's rated values.

#412 · Maintenance and Service

What does an ohmmeter reading indicate if a compressor winding is shorted?

Answer

A short winding shows an abnormally low resistance reading between two terminals compared to rated values. This usually results from insulation breakdown due to overheating inside the winding. It is a condemning fault for that motor winding.

#413 · Maintenance and Service

What does an ohmmeter reading indicate if a compressor winding is open?

Answer

An open winding shows infinite resistance, meaning no continuity, between two terminals that should read a specific value. This is most often caused by a burned-out winding or a blown overload that has not reset. The overload should be ruled out before condemning the winding itself.

#414 · Maintenance and Service

How is a grounded compressor winding tested, and what does any continuity mean?

Answer

One meter lead is placed on any terminal and the other on the casing or a ground lug. Any continuity means the winding insulation has failed and is contacting the shell. This is a condemning fault on virtually every compressor.

#415 · Maintenance and Service

A reciprocating compressor's motor runs and draws roughly normal amperage, but cooling capacity is poor. What mechanical fault should be suspected?

Answer

This points to mechanical valve failure, which shows reduced or no compression differential between suction and discharge even though the motor itself runs normally. It is most common on reciprocating compressor designs. A compression test is used to confirm the diagnosis.

#416 · Maintenance and Service

How does a compression test distinguish healthy compressor valves from worn or broken valve reeds?

Answer

With the system isolated, a compressor with healthy valves pulls a strong vacuum on the suction side and builds discharge pressure quickly. A compressor with worn or broken valve reeds shows suction and discharge pressures pulling toward each other instead of separating. This happens because refrigerant leaks back past the damaged valve on each stroke.

#417 · Maintenance and Service

What does a knocking or rattling noise from a running compressor typically indicate?

Answer

A knocking or rattling noise often indicates a broken or loose internal component, a worn connecting rod bearing, or liquid slugging from flood-back. The distinction usually requires checking suction line temperature and superheat to rule flood-back in or out. All three causes warrant shutting the unit down for further diagnosis.

#418 · Maintenance and Service

What does a high-pitched whining or squealing noise under load suggest on a reciprocating versus a scroll compressor?

Answer

On either design it can point to a worn bearing. On a scroll compressor specifically it can also indicate worn scroll surfaces from lost lubrication or contamination. The noise pattern alone does not fully separate the two causes without further teardown or history review.

#419 · Maintenance and Service

What does a hissing sound while a compressor is running indicate?

Answer

A hissing sound while running can indicate an internal bypass or a leaking discharge valve. Both allow high side gas to leak back internally rather than being fully delivered to the discharge line. This reduces capacity even without an obvious external symptom.

#420 · Maintenance and Service

A compressor hums but does not start, then trips its overload. What causes should a technician suspect?

Answer

This usually indicates a failed start capacitor or start relay that is not bringing the start winding into the circuit. On a scroll compressor it can also mean the system was not equalized before restart, forcing the compressor to start against high head pressure. Both causes prevent the rotor from breaking away, so the motor draws locked rotor current until the overload trips.

#421 • Maintenance and Service

How does an electronic leak detector locate a refrigerant leak in the field?

Answer

It uses a heated or infrared sensor that responds to halogenated refrigerant molecules in the air. These detectors are highly sensitive but also sensitive to airflow and contamination. The technician moves the probe slowly, close to the suspected joint, and confirms any alarm with a second pass after clearing drafts.

#422 • Maintenance and Service

When is the soap bubble test most useful for confirming a refrigerant leak location?

Answer

It is most useful once the general area has been narrowed down, particularly during a standing pressure test with nitrogen on an as-yet uncharged system. A leak forms a visible bubble at the joint, giving precise visual confirmation an electronic sniffer alone cannot always provide. It works especially well on larger or hard-to-access fittings.

#423 • Maintenance and Service

What makes UV dye detection effective for slow leaks that an electronic sniffer struggles to find?

Answer

Fluorescent dye is injected into the system, either at charging time or added later, and suspected areas are inspected with a UV lamp under black light. Any leak site shows a visible dye stain. This is effective for slow leaks that would take too long to build enough concentration for a sniffer, and for leaks hard to sniff directly, such as inside a coil bank.

#424 • Maintenance and Service

What is the correct repair and verification sequence after a refrigerant leak is located and fixed?

Answer

A leaking flare or mechanical joint may only need re-torquing or a new flare, while a pinhole leak in tubing or a coil requires brazing. Any repair on a system that has held refrigerant should then be followed by a standing pressure test with nitrogen, then evacuation, then recharge. It should never simply be topped up over an unconfirmed repair.

#425 • Maintenance and Service

Why is refrigerant leak diagnosis both a technical and a regulatory obligation in Canada?

Answer

Every refrigeration circuit is designed to be sealed for life, so any measurable loss of charge indicates a leak that must be located and repaired, not just topped up. Canadian federal and provincial regulations require leaks above certain thresholds to be repaired and documented. This makes proper diagnosis and record keeping a compliance requirement, not just good practice.

#426 • Maintenance and Service

What symptoms does a run capacitor that has failed weak produce on a compressor or fan motor?

Answer

A run capacitor that has failed weak shows reduced motor performance and higher than normal running amperage. Sometimes the motor overheats over time as a result. The failure is gradual, so amperage trending over several visits can catch it before a hard failure.

#427 • Maintenance and Service

What capacitance meter tolerance is used to judge whether a run capacitor or a start capacitor is still good?

Answer

A run capacitor should read within about six percent of its rated microfarad value. A start capacitor should read within about ten percent for most designs. A reading outside these tolerances means the capacitor should be condemned and replaced.

#428 • Maintenance and Service

Besides a low capacitance reading, what physical signs condemn a capacitor on sight?

Answer

A capacitor that is visibly bulged, split, or leaking oil is condemned regardless of what a capacitance meter reads. Replacement must match the original rating and voltage class exactly. Using a mismatched capacitor can damage the motor it is protecting.

#429 · Maintenance and Service

What causes contactor pitting, and what symptoms does it produce?

Answer

Pitting is a mechanical wear problem caused by repeated arcing across the contact points every cycle. Pitted contacts show visible burning or roughness and can cause intermittent operation or heat buildup from increased resistance. It is a normal wear pattern that worsens gradually over the contactor's life.

#430 · Maintenance and Service

Why is a welded contactor considered a safety hazard, and what should a technician check after replacing one?

Answer

A welded contactor has fused contacts, so the load stays energized even with the coil de-energized, meaning the compressor or fan will not shut off with the thermostat. Any welded contactor is replaced immediately. The technician should also check for the underlying cause, often a locked rotor condition or severe short cycling that caused repeated high-current arcing.

#431 • Maintenance and Service

What are the two distinct ways a failed start relay can affect a single-phase compressor?

Answer

A failed start relay may prevent the start capacitor from dropping out of the circuit, overheating the capacitor and causing it to fail too. It may instead fail to bring the start winding into the circuit at all, leaving the compressor humming without starting. Diagnosis checks relay contacts for continuity in both the energized and de-energized state, plus pickup and dropout voltage against specification.

#432 • Maintenance and Service

What symptoms result when a thermostatic expansion valve is stuck open?

Answer

A TXV stuck open floods the evaporator with too much refrigerant regardless of load. The symptoms are identical to the flood-back signature: low superheat, a cold suction line well past the coil, and possible liquid slugging risk. The valve itself, not the charge, is the root cause here.

#433 • Maintenance and Service

What symptoms result when a thermostatic expansion valve is stuck closed or nearly closed?

Answer

A TXV stuck closed produces the opposite of a stuck-open valve: high superheat, a starved evaporator, and reduced capacity. Frost may be limited to the coil inlet rather than spread evenly across the coil. This is the metering-device version of a starved evaporator fault.

#434 • Maintenance and Service

How does a technician confirm that a TXV is mechanically stuck rather than simply misadjusted?

Answer

The technician checks the bulb charge and mounting, and checks for a plugged external equalizer line if the valve uses one. Then the adjustment is turned while observing whether superheat responds at all. Zero response to adjustment means the valve is mechanically stuck, not simply misadjusted.

#435 · Maintenance and Service

A compressor shuts down under what appear to be normal operating pressures. How is a faulty high pressure switch confirmed as the cause?

Answer

A faulty high pressure switch trips prematurely, below its rated cutout, and is often mistaken for a genuine high head pressure fault. The technician confirms this by gauging actual discharge pressure at the moment of trip and comparing it against the switch's stamped setting. If actual pressure is well below the stamped cutout, the switch itself is at fault.

#436 · Maintenance and Service

Why might a low pressure switch trip during a normal startup pulldown even though the system charge is correct?

Answer

A faulty low pressure switch behaves the same way as a faulty high pressure switch, tripping prematurely on the suction side. This can happen on a normal pulldown dip at startup if the switch's differential is set too tight or the switch has drifted out of calibration. Confirming this requires comparing actual suction pressure at trip against the switch's stamped setting.

#437 · Maintenance and Service

How does thermostat calibration drift show up in the field, and how is it confirmed?

Answer

The thermostat's actual switching temperature moves away from its displayed setpoint over time, often from a failing sensor or a worn bimetal element on mechanical thermostats. It shows up as a space running consistently warmer or cooler than the display suggests. It is confirmed by comparing a calibrated reference thermometer near the sensor against the displayed reading over a full cycle.

#438 · Maintenance and Service

How does scale buildup inside a water-to-refrigerant heat exchanger affect a water-source or geothermal system?

Answer

Scale buildup reduces heat transfer the same way a dirty coil does on an air-source unit. Hard water regions need periodic acid flushing or chemical descaling on a schedule set by local water hardness. Left unaddressed, scale gradually reduces capacity and increases energy use.

#439 · Maintenance and Service

Why must loop flow rate on a water-source or geothermal system be verified against manufacturer specification?

Answer

Low flow reduces heat transfer and can trip a low temperature safety, shutting the unit down. Excessive flow wastes pump energy without improving performance. A flow meter or a temperature-differential calculation across the heat exchanger confirms actual flow against design.

#440 · Maintenance and Service

What happens if antifreeze concentration in a closed-loop geothermal system drifts too low or too high?

Answer

Concentration too low risks freezing and rupturing the loop piping. Concentration too high reduces heat transfer efficiency and increases pumping energy. Concentration should be tested periodically since it can drift from leaks, dilution, or evaporation at fill points.

#441 · Maintenance and Service

A heat pump blows cold air indoors for an extended period and never returns to heating mode. What defrost fault does this suggest?

Answer

This suggests the unit is stuck in defrost, usually from a defrost termination sensor that has failed open or misreads a warmer than actual temperature, or a control board that fails to time out the cycle. The indoor coil acts as the condenser during defrost, so cold air blows indoors until the cycle ends or auxiliary heat covers the load. A failed auxiliary stage on top of a stuck defrost leaves the space cold.

#442 · Maintenance and Service

A heat pump's outdoor coil shows heavy visible frost or ice, but the unit never enters defrost. What should a technician check?

Answer

Likely causes are a failed defrost initiation sensor, a defrost timer or board that fails to trigger its scheduled check, or a reversing valve that has failed and cannot shift into defrost mode even when commanded. The technician confirms reversing valve function by manually forcing defrost through the control board's test function. A proper shift shows the expected pressure and temperature change across the valve.

#443 • Maintenance and Service

A technician hears compressor knocking and finds a cold sweating suction line past the inlet, plus oil foaming in the sight glass. What is happening?

Answer

This is refrigerant flood-back causing liquid slugging inside the compressor, which risks bearing and valve damage. The technician should check for an overcharge, a metering device stuck open, a failed or missing crankcase heater, or very low load conditions. The unit should be shut down and the cause corrected before the compressor is damaged further.

#444 • Maintenance and Service

What federal law governs refrigerant handling in Canada, and which agency enforces it?

Answer

Refrigerant handling falls under the Canadian Environmental Protection Act, 1999 (CEPA). Environment and Climate Change Canada (ECCC) administers CEPA and enforces the Ozone-depleting Substances and Halocarbon Alternatives Regulations under its authority. These regulations cover CFCs, HCFCs like R-22, and the HFC blends and halocarbon alternatives that replaced them.

#445 • Maintenance and Service

State the core prohibition under ECCC's Ozone-depleting Substances and Halocarbon Alternatives Regulations.

Answer

Deliberate release of a controlled refrigerant to the atmosphere, called venting, is prohibited. Recovery is required whenever a refrigeration or air conditioning system is serviced, repaired, or decommissioned, regardless of system size or refrigerant amount.

#446 • Maintenance and Service

Besides opening a valve on purpose, what other actions count as prohibited venting?

Answer

Avoidable losses from careless practice also count as venting under whichever regulation applies to the job, provincial for most private-sector work or the Federal Halocarbon Regulations for federal and Indigenous land equipment. Examples include disconnecting a charged hose without recovering the refrigerant trapped inside it, and abandoning a system with a known leak instead of isolating and repairing it.

#447 • Maintenance and Service

Who can be held liable when a technician vents refrigerant, even by accident through negligent practice?

Answer

Both the technician and the employer can be held liable. The technician can face individual liability for the negligent practice, and the employer can face separate penalties for failing to maintain compliant procedures. These two responsibilities run side by side rather than one substituting for the other.

#448 • Maintenance and Service

Besides the venting prohibition itself, what other technical requirements does ECCC set for recovery work?

Answer

ECCC's regulations also set technical requirements for recovery equipment, minimum recovery efficiency levels for different circuit configurations, and record keeping duties. A technician who knows only the venting prohibition and not this procedural detail will miss exam questions on the mechanics of compliance.

#449 • Maintenance and Service

Why does the recovery requirement apply even on a very small refrigerant charge?

Answer

The applicable regulation, whether the provincial rule covering most private-sector work or the Federal Halocarbon Regulations for federal and Indigenous land equipment, requires recovery whenever a system is serviced, repaired, or decommissioned, regardless of system size or refrigerant amount. The federal Ozone-depleting Substances and Halocarbon Alternatives Regulations instead control manufacture, import, and export nationally. There is no minimum charge threshold that exempts a job from the recovery duty.

#450 • Maintenance and Service

Which international treaty is the source of Canada's domestic refrigerant venting and recovery rules?

Answer

Canada is a party to the Montreal Protocol on Substances that Deplete the Ozone Layer. Canada implemented its Montreal Protocol commitments through CEPA and the Ozone-depleting Substances and Halocarbon Alternatives Regulations.

#451 · Maintenance and Service

What substances did the original Montreal Protocol target, and on what basis?

Answer

The original Montreal Protocol targeted ozone depleting substances, chlorofluorocarbons (CFCs) and later hydrochlorofluorocarbons (HCFCs). It set a phase-down and eventual phase-out schedule for their production and consumption among member countries, based on their ozone-depleting impact.

#452 · Maintenance and Service

What is the Kigali Amendment to the Montreal Protocol, and when was it adopted?

Answer

The Kigali Amendment, adopted in 2016, extended the Montreal Protocol's reach to cover hydrofluorocarbons (HFCs). HFCs do not deplete the ozone layer, but many common HFC blends carry a high Global Warming Potential, so Kigali targets their climate impact instead. Canada ratified the Kigali Amendment and is bound by its schedule.

#453 · Maintenance and Service

Is the Kigali Amendment an instant ban on HFCs, or does it work differently?

Answer

The Kigali Amendment is a phase-down mechanism, not an instant ban. Each party country gets a baseline level of HFC production and consumption expressed in CO₂-equivalent terms, and that allowance is reduced in a series of steps over successive years, shrinking the total legal HFC volume.

#454 · Maintenance and Service

What practical effect does the Kigali HFC phase-down have on refrigerants like R-404A and R-410A?

Answer

The phase-down causes a steady, ongoing reduction in legal supply and a corresponding rise in cost for high GWP HFC refrigerants such as R-404A and R-410A. Availability of newer lower GWP alternatives, including HFO blends and A2L refrigerants, grows in parallel.

#455 • Maintenance and Service

Should a journeyperson expect a single cutover date away from legacy high GWP refrigerants?

Answer

No. The Kigali phase-down is a gradual, multi-year process, so a journeyperson should expect to service a mixed fleet of legacy high GWP systems and newer low GWP systems for a long transition period rather than one cutover date.

#456 • Maintenance and Service

What class of refrigerant is R-22, and what is its current production status in Canada?

Answer

R-22 is an HCFC, and HCFCs were targeted by the original Montreal Protocol schedule for developed countries like Canada. New production and import of R-22 for use in Canada has already been phased out. This is a settled fact, not a future event.

#457 • Maintenance and Service

Explain why the R-22 phase-out should not be confused with the Kigali HFC phase-down.

Answer

The R-22 phase-out targets an HCFC under the original Montreal Protocol schedule, which is already complete for new production and import. The Kigali Amendment is a separate, later mechanism that targets a different chemical family, HFCs, through an ongoing phase-down rather than a completed phase-out.

#458 • Maintenance and Service

What are the only legal sources of R-22 for a system that still needs servicing?

Answer

The only legal sources are recovered R-22 reprocessed to at least the recycled standard, or reclaimed R-22 processed to a purity specification equivalent to new product. No newly manufactured R-22 may be imported or produced for use in Canada.

#459 · Maintenance and Service

Why should recovered R-22 be kept in cylinders segregated from other refrigerant streams?

Answer

R-22 supply is fixed and shrinking, so recovered R-22 has real resale value rather than being scrap. Keeping R-22 recovery cylinders separated from other refrigerant streams keeps the material uncontaminated and reclaimable, protecting its value and its usability.

#460 · Maintenance and Service

Besides R-22, name two CFC refrigerants that are further along the same phase-out path.

Answer

R-12 and R-502 are CFC refrigerants further along the same phase-out path as R-22. They are essentially unavailable except through recovery, recycling, and reclamation channels, since new production has already stopped for the whole ozone depleting substance family on its fixed schedule.

#461 • Maintenance and Service

How is environmental protection jurisdiction for refrigerant handling divided in Canada?

Answer

Environmental protection is a shared jurisdiction, and for refrigerant recovery and venting the provinces do most of the actual work. The federal Ozone-depleting Substances and Halocarbon Alternatives Regulations control manufacture, import, and export nationally, and the separate Federal Halocarbon Regulations cover venting and recovery only for federal and Indigenous land equipment. For nearly all private-sector jobs, each province and territory enacts and enforces its own halocarbon or environmental protection regulation requiring recovery and prohibiting venting.

#462 • Maintenance and Service

What does holding trade certification, provincially or through the Red Seal program, actually authorize?

Answer

Trade certification authorizes a technician personally to perform refrigerant handling work at a competent, trained standard. It demonstrates training in recovery procedures, cylinder handling, and the regulatory obligations that govern refrigerant work, and it operates alongside the employer's own compliance program rather than replacing it.

#463 · Maintenance and Service

List employer-side responsibilities that must accompany a certified technician's compliance work.

Answer

The employer must maintain certified recovery equipment, provide correctly rated and labelled cylinders, keep the shop's refrigerant purchase and disposal records, and ensure technicians actually follow their trained procedures. A fully certified technician can still work in a non-compliant shop if the equipment is unmaintained or paperwork is not kept.

#464 · Maintenance and Service

Define recovery in the context of refrigerant handling.

Answer

Recovery is the act of removing refrigerant from a system, in any condition, and storing it in an external container without necessarily testing or processing it. Recovered refrigerant may still contain moisture, acid, oil, or non-condensable gases.

#465 • Maintenance and Service

Why is recovery described as the mandatory first step common to every refrigerant handling path?

Answer

Every time a system is opened for service, refrigerant must first be removed and stored externally before any further processing decision is made. Whether the gas is later recycled, reclaimed, or otherwise handled, recovery always happens first.

#466 • Maintenance and Service

Define recycling as it applies to recovered refrigerant, and describe the equipment typically used.

Answer

Recycling reduces contamination in recovered refrigerant through oil separation, moisture removal with a filter drier, and single or multiple pass filtration. It is usually performed with equipment on site at the job or in the shop, rather than at a specialized facility.

#467 · Maintenance and Service

What is the reuse limitation on recycled refrigerant compared to reclaimed refrigerant?

Answer

Recycled refrigerant is cleaned enough to be reused in the same system, or in other equipment owned by the same customer, but it does not meet the full purity standard required for resale as new refrigerant. Recycling is a lower bar meant for closed loop reuse, not general resale.

#468 · Maintenance and Service

Define reclamation and the purity standard it must meet.

Answer

Reclamation is a full reprocessing of used refrigerant to the same purity specification as newly manufactured product, verified against a recognized chemical purity standard. It requires specialized equipment, typically a dedicated reclamation facility rather than field or shop equipment.

#469 · Maintenance and Service

Why can reclaimed refrigerant legally be resold into a different customer's system, unlike recycled refrigerant?

Answer

Reclaimed refrigerant has been fully reprocessed to the same purity specification as new product and verified against a recognized purity standard. That full interchangeability lets it be legally resold into any compatible system, including systems belonging to other customers, unlike recycled refrigerant.

#470 · Maintenance and Service

How does reclamation keep a phased out refrigerant like R-22 available for ongoing service?

Answer

Reclaimed material is treated as equivalent to new product for regulatory purposes because it meets the same purity specification. This lets reclaimed R-22 legally substitute for new production, which is otherwise unavailable, keeping existing R-22 systems serviceable.

#471 • Maintenance and Service

What is a recovery machine, and how does its function differ from the refrigeration system itself?

Answer

A recovery machine is a dedicated compressor and condenser package built to pull refrigerant out of a customer's system and push it into a separate recovery cylinder. It removes and stores refrigerant rather than circulating it the way the system's own compressor does.

#472 • Maintenance and Service

Compare vapour recovery and liquid recovery, and explain why liquid recovery is faster on large charges.

Answer

Vapour recovery draws off refrigerant vapour and condenses it inside the recovery machine, while liquid recovery removes refrigerant in its liquid state. Liquid recovery is faster on systems holding a larger charge because liquid removal moves far more mass per minute than vapour removal.

#473 · Maintenance and Service

Before starting a recovery machine, what must a technician confirm about the recovery cylinder?

Answer

The technician connects the recovery machine between the system's service ports and an evacuated, correctly rated, properly labelled recovery cylinder. Gauges must confirm the cylinder is not already full past its allowable fill limit before recovery begins.

#474 · Maintenance and Service

How should a technician judge when a recovery is actually complete?

Answer

The technician watches system side pressure and pulls down to a specified vacuum level appropriate to the refrigerant and equipment type, rather than stopping on a timer. A system left with residual refrigerant pressure after recovery is not actually recovered.

#475 · Maintenance and Service

Why is refrigerant left in hoses before disconnection a compliance and safety concern?

Answer

Refrigerant remaining in hoses after recovery can be vented by accident when a fitting is cracked open. A careless technician who disconnects a charged hose without recovering the trapped refrigerant is committing a form of prohibited venting.

#476 · Maintenance and Service

Describe push-pull recovery and the type of system where it is most commonly used.

Answer

Push-pull recovery uses the recovery machine's own compressor to pull vapour off the recovery cylinder and push it into the system, forcing liquid refrigerant out of the system's liquid receiver and into the recovery cylinder under the resulting pressure differential. It is common on larger commercial systems, where straight vapour or liquid recovery alone would take too long.

#477 · Maintenance and Service

After recovering refrigerant from a system, what should a technician confirm the quantity against?

Answer

The technician must weigh or otherwise confirm how much refrigerant was recovered against the equipment's nameplate charge. This verifies the recovery was complete and provides accurate data for the compliance documentation the shop must keep.

#478 · Maintenance and Service

Are recovery cylinders treated as disposable containers under transport and safety rules?

Answer

No. Recovery cylinders are refillable pressure vessels, treated as such under transport and pressure vessel safety rules. They are reused across many recovery jobs rather than discarded after one fill.

#479 • Maintenance and Service

Describe the conventional colour scheme for a North American recovery cylinder and how it differs from a new disposable cylinder.

Answer

A recovery cylinder is conventionally painted grey with a yellow shoulder or top. New, disposable refrigerant-specific cylinders use different colours, for example R-410A cylinders are typically rose-coloured and R-22 cylinders are light green.

#480 • Maintenance and Service

Since a grey-and-yellow recovery cylinder is a mixed-use container, how must it be identified between different refrigerant uses?

Answer

The cylinder must carry a legible label stating whatever refrigerant it currently holds, rather than being identified by tank colour alone. One recovery cylinder may serve different refrigerants across its service life, provided it is fully evacuated and correctly relabelled between uses.

#481 • Maintenance and Service

Which Canadian regulatory framework governs refillable refrigerant recovery cylinders, and what physical rating do they carry?

Answer

Refillable recovery cylinders fall under Transport Canada's Transportation of Dangerous Goods (TDG) requirements, which mirror the DOT specification framework used in the United States. The physical cylinder commonly carries a DOT-4BA or DOT-4BW type rating stamped into it, setting its maximum working pressure and safe fill limit. A DOT-39 rating marks a non-refillable, single-trip cylinder and must never be refilled or used as a recovery cylinder.

#482 • Maintenance and Service

How often must a standard DOT-4BA/4BW recovery cylinder undergo hydrostatic requalification testing, and what happens if it is overdue?

Answer

A standard DOT-4BA or DOT-4BW type cylinder typically requires hydrostatic requalification at a five-year interval, stamped with the test date on the cylinder. A cylinder outside its requalification window cannot legally be filled or transported, and using it is both a regulatory violation and a real safety hazard. DOT-39 cylinders are non-refillable, single-trip containers that are never requalified or refilled.

#483 • Maintenance and Service

Describe the letter-and-number safety group system used to classify refrigerants by toxicity and flammability.

Answer

A letter indicates toxicity, A for lower toxicity and B for higher toxicity. A number indicates flammability: 1 means no flame propagation, 2L means mildly flammable, 2 means flammable, and 3 means higher flammability.

#484 • Maintenance and Service

What safety classification do R-410A and R-22 carry, and what does that classification mean?

Answer

R-410A and R-22 are classified A1. That means both refrigerants have lower toxicity and show no flame propagation under standard test conditions.

#485 • Maintenance and Service

Name two A2L refrigerants that have entered common use under the Kigali low GWP shift.

Answer

R-32 and R-454B are A2L refrigerants that have entered common use. Their adoption is part of the industry-wide shift toward lower GWP options driven by the Kigali Amendment's HFC phase-down.

#486 • Maintenance and Service

What does the 2L designation mean for the burning behaviour of a refrigerant like R-32?

Answer

The 2L designation means the refrigerant is mildly flammable. It will burn under specific conditions, but it has a low burning velocity and requires a higher concentration in air plus a real ignition source to sustain combustion.

#487 • Maintenance and Service

How does an A2L refrigerant's risk category compare to a fully flammable A2 or A3 refrigerant such as propane (R-290)?

Answer

A2L refrigerants sit in a distinct, lower-risk category than fully flammable A2 or A3 refrigerants like propane (R-290). Mild flammability under specific conditions is a lower hazard than the fuller flammability of the A2 or A3 class.

#488 • Maintenance and Service

What must be true of recovery machines and leak detectors used on an A2L refrigerant system?

Answer

Service equipment used on A2L systems must be rated for mildly flammable refrigerants. Standard A1-only equipment lacks the spark-free components and enclosure ratings needed to safely handle a refrigerant that can ignite.

#489 · Maintenance and Service

What storage and ventilation factor becomes important with A2L refrigerants that is less relevant with A1 refrigerants?

Answer

Storage areas and shop ventilation must account for a refrigerant accumulation reaching a flammable concentration in an enclosed space. This matters particularly at floor level, since A2L refrigerant is heavier than air and can pool there.

#490 · Maintenance and Service

What must a technician do before brazing on a charged A2L refrigerant line?

Answer

The line requires full evacuation and purging before any brazing or other hot work. Brazing directly on a charged A2L line, something a technician might do without a second thought on an A1 system, introduces a real ignition risk.

#491 • Maintenance and Service

What design consideration applies to electrical enclosures near a possible A2L leak point?

Answer

Electrical enclosures and other system components near a possible A2L leak point should carry flammable-refrigerant-rated designs from the manufacturer. This reduces the chance of an ignition source being present where a mildly flammable refrigerant could accumulate.

#492 • Maintenance and Service

How is a system losing refrigerant at an elevated rate over a service interval treated under this framework?

Answer

It is treated as a compliance problem, not just a performance problem. A leak rate above the threshold that applies to that system's size and type calls for repairing the leak and confirming the repair, rather than simply topping up the charge.

#493 • Maintenance and Service

Why does repeatedly recharging a leaking system without repairing it defeat the purpose of the recovery and venting rules?

Answer

Every top up on an unrepaired leaking system effectively feeds a slow, unauthorized release to atmosphere. This undermines the entire purpose of the recovery and venting rules even though no single event looks like deliberate venting.

#494 • Maintenance and Service

How must refrigerant that cannot be recycled or reclaimed, and contaminated recovered oil, be handled for disposal?

Answer

Both must be handled as regulated waste rather than disposed of through ordinary channels. This means using a licensed waste handler or refrigerant reclaimer, keeping material properly labelled and contained until pickup, and never disposing of either down a drain or into general waste.

#495 • Maintenance and Service

What records should a compliant service company keep to demonstrate refrigerant accountability?

Answer

Records should cover refrigerant purchases, the quantity recovered from each job matched against the system's original charge, refrigerant sent for recycling or reclamation, and the final disposition of any refrigerant or cylinder leaving the shop. These records show refrigerant is accounted for with no unexplained gap suggesting undocumented venting.

#496 • Maintenance and Service

What specific information should a technician record on a service ticket to support the shop's compliance record?

Answer

A technician should record cylinder weight in and out, the refrigerant type, and the job reference on every service ticket. This front-line record keeping is what makes the shop's overall compliance documentation possible.

#497 • Maintenance and Service

Who is the intended audience for a service company's refrigerant record keeping, and what gap are the records meant to rule out?

Answer

An inspector or auditor is exactly who will ask to see these records. Complete records rule out an unexplained gap between refrigerant purchased and refrigerant accounted for, which would otherwise suggest undocumented venting.

#498 • Maintenance and Service

Why must every independent refrigeration circuit be recovered before decommissioning a piece of equipment?

Answer

Decommissioning carries the same recovery obligation as any other service call, and the full refrigerant charge must be recovered from every circuit before cutting, crushing, or disposal. Some equipment has multiple independent refrigeration circuits or a pump-down receiver holding charge separately from the rest of the system.

#499 • Maintenance and Service

During decommissioning, what should a technician do if the recovered quantity falls significantly short of the nameplate charge?

Answer

The technician should confirm the recovered quantity against the equipment's nameplate charge where possible and investigate any significant shortfall, rather than assuming the missing refrigerant simply was not there. A large shortfall can indicate a long-standing undetected leak that itself needs documenting.

#500 • Maintenance and Service

What must happen before a decommissioned system moves on to demolition or scrap processing, and how should its compressor oil be handled?

Answer

Once recovery is confirmed complete, the equipment can be tagged refrigerant-free before demolition, recycling, or scrap processing, protecting downstream workers from handling a pressurized system as inert scrap metal. Compressor oil should be drained and handled as contaminated waste, since it typically retains dissolved refrigerant.